MIL-B-18796E
9 May 1984
SUPERSEDING
MIL-B-18796D
6 August 1976

MILITARY SPECIFICATION

BURNERS, SINGLE: OIL, GAS, AND GAS-OIL COMBINATION (400,000 BTU'S PER HOUR AND OVER INPUT CAPACITY)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers single oil, gas, and gas-oil combination power burners designed for installation in a single burner boiler or furnace. The burner unit shall include all components and accessories for the installation (see 6.1).

1.2 Classification. Power burners shall be one of the following sizes, classes, control sequences, and combustion control systems, as specified (see 6.2 and 6.6).

Sizes - Input Capacity.

Size 1 - 400,000 to 2,500,000 British thermal units per hour (Btu/hr) input capacity.
Size 2 - 2,500,001 to 5,000,000 Btu/hr input capacity.
Size 3 - 5,000,001 to 12,500,000 Btu/hr input capacity.
Size 4 - 12,500,001 to 30,000,000 Btu/hr input capacity.
Size 5 - 30,000,001 Btu/hr input capacity and above.

Classes - Type of Fuels (see 3.8, 3.12, and 6.1.2).

Class 1 - Heavy oil-fired (No. 4, No. 5, No. 6, or Navy special as specified).
Class 2 - Light oil-fired (No. 2, diesel, and JP-5 as specified).
Class 3 - Gas fired (with commercial name of gas as specified).
Class 4 - Combination heavy oil and gas-fired (grade of oil and commercial name of gas as specified).
Class 5 - Combination light oil and gas-fired (grade of oil and commercial name of gas as specified).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4530

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Class 6 - Combination light oil, heavy oil, and gas fired (grade of oils and commercial name of gas as specified)(see 6.1(g)).

Control sequences (see 3.10):

Control sequence IA - Automatic system (recycling) (For size 1 burner only).
Control sequence IB - Automatic system (recycling) (For all burners).
Control sequence II - Automatic system (nonrecycling) (For size 4 and 5 burners only).
Control sequence IV - Supervised manual system with gas-electric igniter, for size 4 and 5 burners only.

Combustion control systems (see 3.11, 6.1.1(d), and table VI):

On-Off.
High-Low-Off.
Positioning (simple).
Parallel positioning with air/fuel ratio control.
Parallel metering with lead-lag capability.
Metering with oxygen compensation.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified (see 6.2), the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

FF-B-171 - Bearings, Ball Annular, (General Purpose).
ZZ-B-225 - Belts, V: Industrial (Multiple Drive).
VV-F-800 - Fuel Oil, Diesel.

MILITARY

MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant, (for Treatment of Communications, Electronic, and Associated Equipment.
MIL-F-859 - Fuel Oil, Burner.
MIL-B-3180 - Boilers and Related Equipment, Packaging of.
MIL-T-5624 - Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-T-13867 - Treatment, Moisture and Fungus Resistant for Fire Control Electrical and Electronic Instruments and Equipment.
MIL-I-24092 - Insulating Varnish, Electrical, Impregnating, Solvent Containing.

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STANDARDS

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for Control of Electromagnetic Interference.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurements of.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specifications to the extent specified herein.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Code of Federal Regulations (CFR).
Title 40, Part 60.344 - Visible Emissions, - Method 9.

(Applications for copies should be addressed to the Public Affairs Office, Environmental Protection Agency, Rockville, MD 20852.)

(Copy of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on the date of invitation for bids or request for proposal shall apply.

AIR MOVING AND CONDITIONING ASSOCIATION (AMCA)

Standard 210 - Test Code for Air Moving Devices.

(Application for copies should be addressed to the Air Moving and Conditioning Association, 205 West Touhy Ave., Park Ridge, IL 60068.)

AMERICAN GAS ASSOCIATION LABORATORIES (AGA)

Directory of Certified Appliances and Accessories.

(Application for copies should be addressed to the American Gas Association Laboratories, 8501 East Pleasant Valley Road, Cleveland, OH 44131, or 1425 Grande Vista Ave., Los Angeles, CA 90023.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B15.1 - Safety Code for Mechanical Power-Transmission Apparatus.
Z83.3 - Gas Utilization-Equipment in Large Boilers.
CSD-1 - Controls and Safety Devices for Automatically Fired Boilers.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D396 - Fuel Oils.
D2156 - Test for Smoke Density in the Flue Gases from Distillate Fuels.

(Applications for copies should be addressed to the American Society for Testing an Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section 8.
Pressure Vessels, Division 1.
PTC-4.1 Steam Generating Units.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street New York, NY 10017.)

FACTORY MUTUAL ENGINEERING CORPORATION (FM)

Factory Mutual Approval Guide.

(Application for copies should be addressed to the Factory Mutual Engineering Corporation, Factory Mutual System, 1151 Boston-Providence Turnpike, Norwood, MA 02062.)

INSTRUMENT SOCIETY OF AMERICA (ISA)

S18.1 - Standard for Annunciator Sequences and Specifications.

(Application for copies should be addressed to the Instrument Society of America, P.O. Box 12277, Research Triangle Park, NC 27709.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ICS 1 - General Standards for Industrial Controls and Systems.
ICS 2 - Standards for Industrial Control Devices, Controllers, and Assemblies.
ICS 6 - Enclosures for Industrial Control Systems.
MG 1 - Standards for Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street N.W., Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 54 - National Fuel Gas Code.
No. 58 - Liquefied Petroleum Gases, Storage and Handling.
No. 70 - National Electrical Code.
No. 85A - Prevention of Furnace Explosions in Fuel Oil and Natural Gas-fired Single Boiler Burner Furnaces.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

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RUBBER MANUFACTURERS ASSOCIATION (RMA)

Engineering Standards for Multiple V-Belt Drives.

(Application for copies should be addressed to the Rubber Manufacturers Association, 444 Madison Avenue, New York, NY 10022.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

UNDERWRITERS LABORATORIES INC. (UL)

UL296 - Oil Burners.
UL353 - Limit Controls.
UL795 - Commercial Gas Heating Equipment.

(Application for copies should be addressed to Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The power burner unit (see 6.6) shall consist of a complete fuel system, atomizers, nozzles, atomization means, forced or induced draft fan (see 3.15), an automatic control system complete with combustion and flame failure controls, firing sequence programmer, safety interlocks, limit controls, control cabinets or panels, trim, and appurtenances pertinent to burners as specified herein. All units shall be factory-wired and assembled as practical except for readily installed appurtenances, such as pressure and temperature gages, oil heater piping, and wiring of the burners to the control panel. The burners shall be complete and ready for operation when installed on the applicable equipment and connected to the fuel, and electrical supplies.

3.2 First article. When specified (see 6.2), the contractor shall furnish a complete burner for first article inspection and approval (see 4.2.1 and 6.4).

3.3 Standard commercial product. The power burner shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the burner being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market

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through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.4 Standards compliance. Power burners, including any supplementary control devices, safety interlocks, or limit controls required under this specification shall meet the requirements of the following standards as applicable:

a. Oil-fired units - UL296 or NFPA No. 85A.
b. Gas-fired units - UL795, NFPA No. 85A, or ANSI Z83.3 (for power burners).
c. Combination gas and oil-fired units - UL296, UL795, NFPA No. 85A, or ANSI Z83.3 (for power burners).

All controls, not covered by the above shall have the UL label or listing mark, or shall be listed in the Factory Mutual Approval Guide.

3.4.1 Certification. Prior to approval of the sample unit furnished for first article inspection, if such inspection is specified, or prior to the first shipment, if first article inspection is not required, the contractor shall submit evidence satisfactory to the contracting officer or his authorized representative that the units to be furnished under this specification meet the standards cited in 3.4, and the approval requirements of UL and FM for the fuel and combustion controls, combustion safeguard controls, and safety shutoff valves.

3.4.1.1 Burners with rated input under 30 million Btu/hr except for burners used on watertube boilers with rated input above 12.5 million Btu/hr. Acceptable evidence of meeting these standards will be the official UL label, a listing in the AGA Directory of Certified Appliances and Accessories, or a certified test report (see 6.3), from a recognized independent laboratory acceptable to the Government, indicating that the burner and components meet the requirements of the published standards cited in 3.4, including the specified methods of testing.

3.4.1.2 Burners with rated input of over 30 million Btu/hr and burners for watertube boilers with rated input above 12.5 million Btu/hr. Acceptable evidence of meeting the specified requirements herein may be a certificate of compliance (see 6.3), from any nationally recognized testing organization equipped and competent to perform such services or a certificate of compliance (see 6.3), executed by an authorized official of the contractor, stating that the items have been tested and that the units meet the requirements of the specified standards in 3.4, including specified methods of testing. Acceptable evidence of meeting the fuel and combustion controls, combustion safeguard controls, and safety shutoff valves will be the official UL label, FM approval mark prescribed in the Approval Guide of the Factory Mutual System, or a certified test report (see 6.3), from a recognized independent laboratory acceptable to the Government, indicating that the components meet the approval requirements of UL or FM.

3.5 Design. The power burners shall be designed in accordance with the applicable requirements of UL296, UL795, NFPA No. 85A, FM, ANSI Z83.3 and any additional requirements specified herein. The burner shall be of a design acceptable for the type of operation specified herein. When the design is not practicable for assembly of specified components, serving as mounted controls,

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such components shall be supplied as part of the burner unit for installation by others. The design shall meet the performance requirements of 3.8, 3.8.1, and 3.8.2. The equipment design and accessory installations shall permit accessibility for maintenance and service in the field. The design shall be such as to prevent conditions hazardous to personnel or deleterious to equipment.

3.5.1 Design instructions. When the burner is intended for conversion or replacement of the existing burner, the burner shall be suitable for the installation as specified (see 6.2, 6.3, and 6.5).

3.6 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.7 Materials. Materials shall be as specified herein. Parts for which materials are not definitely specified shall be fabricated of materials acceptable under referenced UL standards or for FM approved equipment (see 6.6), and, if not covered therein, of materials used in the manufacturer's standard product. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.8 Performance. Unless otherwise specified (see 6.1.2 and 6.2), the performance requirements shall be as follows:

a. Fuel oil (percent by weight, as fired).

Ultimate analysis:

Hydrogen	10.33
Carbon	87.87
Oxygen	0.25
Nitrogen	0.14
Sulfur	1.16
Ash	0.25

Higher heating value - - 18,400 British thermal units per pound (Btu/lb)

b. Natural gas (percent by volume, as fired).

Ultimate analysis:

Methane	90.0
Ethane	5.0

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Nitrogen	5.0
Ash	Nil
Moisture	0.01

Higher heating value - - 1,075 British thermal units per cubic foot (Btu/cu ft) or (21,824 Btu/lb)

The gas supply pressure at the site upstream from the service pressure regulator shall be as specified in the contract (see 6.2). When burners are acquired for stock and the conditions at the site are not known in advance, a gas pressure at the inlet to the gas piping of 2 pounds-force per square inch gage (psig) shall be assumed for design purposes.

3.8.1 Capacity. When the burner is fired at maximum continuous capacity, the input, expressed in Btu/hr, shall be as specified (see 6.2 and 6.5).

3.8.2 Combustion efficiency. The combustion efficiency, as determined by flue-gas analysis for carbon dioxide(CO_2) and carbon monoxide for fuels specified herein shall be as shown in table I. The combustion efficiency for modulating units shall be not less than the average of the three rated loads of table I. For on-off and high-low-off controlled units, combustion efficiency shall not be less than the average of three maximum and minimum firings at rated loads. When fired with fuels specified herein, the burner shall not exceed the smoke density readings listed in table I when measured in accordance with ASTM D2156 (Shell-Bacharach) of the values except for short periods not to exceed 3 minutes. When specified (see 6.2), the burner shall be designed to burn the fuels identified herein, so that the particulate, nitrogen oxide, and sulfur dioxide emissions as specified (see 6.1.1(k) and 6.2) shall not exceed the limits when determined in accordance with 4.5.4. Both stable firing and combustion efficiency shall be maintained over the entire firing range required by the burner turndown (see 3.12, 3.12.1 and 3.12.2).

3.9 Control systems. Each burner shall be equipped with the following control systems:

a. Solid state programming control system (see 3.10): In conjunction with the electronic flame-failure control, safety interlocks and limit switches, the programming control system shall govern the prefiring, light-off, and shutdown cycle.
b. Combustion control system (see 3.11): This system shall automatically regulate the firing cycle in accordance with the load demand.

3.10 Programming control systems. The control sequences of the programming control systems shall be as specified in 1.2, as tabulated in table II, and as defined in NFPA No. 85A. The control sequences for the various sizes of burners along with their pertinent features and reference paragraphs are included. The basic programming and flame-failure controls shall be furnished as an integrated system by the burner manufacturer and shall be designed to function in combination with other system components to insure the functional integrity of the total system. The program control system shall, as a minimum, indicate by indicating lights the following occurrences: activation of power on or start of blower, prepurge cycle, trial for pilot ignition, when main flame is detected, and when operating under

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automatic control. A post-purge period will not be mandatory on size 1 burners, but shall be mandatory on size 2, 3, 4, and 5 burners.

3.10.1 Control sequence IA. Control sequence IA shall be an automatic sequence (recycling) system having direct electric ignition for use with light oil size 1 burners. Burners shall perform the following operations:

a. Manually operated switches shall apply power to the burner and energize the programming control.
b. Fan and timer start a forced-air-change purge of the combustion chamber of the furnace. The minimum purge time shall be not less than 30 seconds and purge air-flow rate shall reach not less than 60 percent of the flow rate at maximum output. For High-Low-Off combustion control, purge time may be 90 seconds when the air-flow rate at low-fire start is not less than one-third maximum. All interlocks and electrical contacts, including a fan interlock switch shall close or be closed before the end of the purge period and have indicated safe operating conditions (see 3.10.6).
c. After completion of the purge, the timer shall actuate the ignition, open the main fuel-safety-shutoff valve, or at the manufacturer's option, the valve opening may be delayed for a stipulated period.
d. When the main fuel valve opens, main burner establishing period shall begin, and activate a green indicating light. If the flame-sensing device does not sense the main flame in the time specified in table II, a flame-failure safety shutdown (see 3.10.5) shall occur.
e. After the main flame is detected, the burner shall fire until the heat demand is satisfied. Upon satisfaction of heat demand, the burner shall shut off in accordance with the manufacturer's design procedure, and shall restart automatically when the demand is again established.
f. The burner shall establish safety shutdown automatically as a result of flame-failure (see 3.10.5), or of limit-switch or fuel and air-flow safety-interlock opening as specified in 3.10.6. The safety shutdown shall accomplish the following in the order listed below and activate an alarm, a red indicating light, and deactivate the green light in table V for control panel.

(1) Shut off fuel supply to main burner.
(2) Interrupt spark ignition if in operation.
(3) Where used, open recirculating valve.
(4) Shut down fan.
(5) Lock out programming control to require manual reset.

3.10.2 Control sequence IB. Control sequence IB shall be the automatic sequence (recycling) system with proved igniter for use on all gas, oil, or combination gas-oil burners. Burners shall perform the following operations:

a. Manually operated switches shall turn on the power to the burner and energize the programming control indicated by a blue indicating light. Switches shall start the heating and circulation of the oil where an auxiliary heater and recirculating system is used.
b. Prefiring cycle of the burner shall accomplish the following sequences:

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(1) On sizes 3, 4, and 5 , prove the fuel safety shutoff valve closed, for gas firing this shall be the downstream valve.
(2) Prove no flame present at the burner.
(3) Start fan.
(4) Satisfy air-flow interlock.
** (5) Where atomizing medium is used and if not already on, admit medium to main burner and on size 3 and larger burners activate a colored indicating light as indicated in table V.
** (6) Where atomizing medium is used, satisfy atomizing medium interlock.
** (7) Satisfy fuel pressure and limit switches and interlocks.
(8) Prove purge air flow by: Satisfying air pressure and "open damper" interlocks for all dampers in the flow path and satisfying all air flow interlocks. Purge air flow rate shall reach not less than 60 percent of the flow rate at maximum output for burner sizes 1 through 3, and not less than 70 percent for size 4 burners and larger.
(9) The purge shall be sufficient for at least four air changes of the combustion chamber, heat exchangers, and flue passages for burner sizes 1 through 3 and eight air changes for size 4 burners and larger. Air flow during the time to open and return the damper to light-off position may be included in computing the time required for the air changes.
(10) Set controls to light-off position (low fire) for High-Low-Off and modulating combustion controlled burners.
(11) On units with High-Low-Off and modulating combustion controls, prove dampers, main flame fuel operational control valves in low-fire and light-off position respectively.

c. Light-off cycle of the burner with the required interrupted igniter and all interlocks satisfied (see 3.10.6) shall accomplish the following sequences in the order listed:

(1) Ignite igniter with ignition system as in 3.13.

(2) Prove igniter flame within the igniter establishing period as specified in table II.

(a) If proven, open the main fuel safety-shutoff valves to admit fuel to the burner. For an oil burner other than the return flow type, the burner shall simultaneously shut off oil recirculating flow.

(b) If not proven, an ignition-failure shutdown shall occur in accordance with 3.10.2(g).

**NOTE: The sequential order of these items is optional, atomizing medium must be dry.

(3) Prove main flame within the maximum time for main-flame establishing period of table II.

(a) If proven, shut off igniter, vent the gas igniter piping, release burner to operation cycle and activate a green light in table V for control panel.

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(b) If not proven, a flame-failure shutdown shall automatically occur in accordance with 3.10.2(g).

d. Operation cycle of the burner shall be accomplished by a combustion control system specified in 3.11.

e. Normal shutdown cycle shall accomplish the following in the order listed.

(1) Shut off fuel to the main burner.
(2) For oil:

(a) Where used, open recirculating valve.
(b) Where used, shut off atomizing medium, if desired.

(3) For gas:

(a) Vent gas piping for size 3 and larger between the main burner safety shut-off valves to atmosphere.

(4) After a post purge on size 2 burners and larger shut down fan, if desired and deactivate green light in table V (see 6.1.1(i)).

f. When the load demand is satisfied, automatic (recycling) burners shall accomplish a recycling shutdown (see 6.6), and the burner shall be allowed to recycle when the load demand is reestablished within the preset operating range.

g. Nonrecycling safety-shutdowns (see 6.6) shall accomplish the following in the order listed and activate an audible alarm and a red indicating light in table V.

(1) Shut off fuel supply to the igniter for ignition failure and main burner for main flame failure.
(2) Interrupt spark to the igniter, if in operation.
(3) Oil:

(a) Where used, open recirculating valve.
(b) Where used, shut off atomizing medium.

(4) Gas: Vent the gas piping to the igniter and to the main burner.

(5) After a post purge on size 2 burners and larger, shut down fan, if desired.

(6) Lockout programming control system (see 3.10.5) to require manual reset.

3.10.3 Control sequence II. Control sequence II shall be the automatic (nonrecycling) (see 6.1.1(a)) system with proved igniter for use on all gas, oil, or combination gas-oil burners. The operating sequence shall include the following:

a. The provisions of subparagraphs (a), (b), (c), (d), (e), and (g) for automatic (recycling) sequence in 3.10.2 shall apply.

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b. When load demand establishes a normal shutdown, the burner shall not recycle. Burners shall require, for restart, the following in the order listed:

(1) Manual actuation of the reset button of the programming control.
(2) Manual actuation of the burner control start switch.
(3) Upon indication of proven igniter in the light-off cycle, open the manual reset safety-shutoff valve to admit fuel to the burner. A light-off time limit of 2 minutes shall be allowed between completion of purge and opening of the manual reset safety-shutoff valve, and is supplemental to and separate from the main-flame establishing period. If the manual reset safety-shutoff valve is not opened in the allotted time, a nonrecycling safety-shutdown shall occur.

3.10.4 Control sequence IV. Control sequence IV (oil) and control sequence IV (gas) shall be the supervised manual system with spark-ignited igniter for use on oil or gas burners over 12.5 million Btu/hr input capacity in accordance with NFPA No. 85A (see 6.1.1(a)). Burners shall perform the specified steps when initiated manually by the operator and the specified interlock functions shall be automatically satisfied and indicated to the operator by position and by an indicating light as specified in 3.14. Control sequence IV, (gas-oil), shall be the supervised manual system with spark-igniter ignition for use on gas-oil combination burners over 12.5 million Btu/hr input capacity. The sequence shall conform to the requirements for gas and oil burners above and all controls shall be interlocked with the manual fuel change-over switch to prevent firing of both fuel simultaneously. The burners shall not be allowed to recycle automatically after any shutdown.

3.10.5 Flame-failure control system. Each burner shall be equipped with a standard (nonproprietary) flame-failure control system (see 3.9 and 6.6). The flame-failure control system shall have a repetitive self-checking circuit and shall consist of a flame-sensing element properly connected to the programming control system for shutdown of the burner in the event of ignition failure or flame failure during the firing cycle. The burner shall not be allowed to recycle after ignition failure or main flame-failure without manual reset. The flame failure reaction time including allowable factory tolerances shall be from 2 to 4 seconds for all burners at rated control voltage in an ambient temperature of 70 Degrees Fahrenheit (°F). Flame-sensing elements shall have solid state amplifiers and shall not respond to ignition spark, hot refractory, or reflection of flame on atomizing media or oil spray. Flame-sensing elements for igniter flames shall detect only flames of sufficient intensity to safely light the main burner. Flame-sensing elements shall be so connected as to prevent startup, if flame is present or if the contacts are stuck in the flame-sensing position. The flame-failure control circuit shall be so designed that the failure of any component prevents restart of the burner after shutdown, and shall incorporate automatic start-check at the beginning of each startup cycle. Flame-sensing elements shall consist of one type, or a combination of the following types within the limitations indicated:

a. Ultraviolet cells may be used for gas or oil flames, but shall not be used for igniter proving unless, (1) the spark is shut off during the last part of the proving period, or (2) the scanner is

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sighted so as to exclude spark detection, or (3) the scanner is equipped with a sight-tube to exclude spark detection.

b. Infrared (lead sulfide) cells may be used for gas or oil flame. Infrared cells shall not be used in locations where the cell could be actuated by residual radiation emanating from hot refractory after flame extinguishment.

3.10.5.1 Repetitive self-checking circuit. The repetitive self-checking circuit shall automatically check all components of the flame-failure system including the flame detector. The component check of the detector is not required if the detector is fail-safe such as lead sulfide cells. During burner operation the self-checking circuit shall simulate failures at intervals not greater than 6 seconds. The occurrence of a flame simulating failure anywhere in the flame-failure system or the cessation of the cyclic operation of the self-checking circuit shall cause a safety shutdown and lockout.

3.10.6 Safety interlocks and limit switches. Each burner shall be equipped with the applicable safety interlocks and limit switches as indicated in table III and table IV. The safety interlocks and limit switches shall sense the limit or off-limit conditions indicated, cause a nonrecycling safety shut- down or recycling shutdown (see 6.6) of the burner in the specific cases annotated in table III, and determine the position of the sensing element switch without disassembly of the device. All nonrecycling shutdowns (see 6.6) shall require a manual reset of the interlock, limit switch, and primary flame safeguard control, and shall be signaled by a red indicating light and an audible alarm. The safety interlocks and limit switches shall conform to UL353. All safety interlocks and limit switches, including boiler and equipment limit switches, shall be wired in the hot side of the burner limit control circuit and not in any external circuits such as the pilot or fuel valve circuits.

3.11 Combustion control systems. Combustion controls specified in 1.2, except On-Off controls shall automatically regulate the firing rate in accordance with the load demand and shall meet the specified burner turndown (see 6.6) in 3.12.1 or 3.12.2, as applicable. Control systems shall maintain air-fuel mixtures within the limits required for stable and efficient combustion throughout the specified firing range and, as applicable, during changes in firing rates The combustion control systems shall maintain the load, as applicable, within the following tolerance limits expressed as a percent of the set point values of the operating control.

(a) On-Off - $\pm$6 percent.
(b) High-Low-Off - $\pm$5 percent.
(c) Modulating - $\pm$3 percent.

The specified tolerances shall apply to any load which, within a 1 minute period, swings from a steady state condition to an increase (or decrease) in load equal to a maximum of 10 percent of the full rated capacity of the burner. Regulation tolerances shall apply to any steady-state conditions within the specified burner turndown. For On-Off systems with subtractive differential settings, the specified tolerance shall be applied as a plus tolerance to the OFF set point and as a minus tolerance to the ON set point. On units with High-Low-Off and modulating controls, the combustion control shall release from the low-fire position and adjust the firing rate in

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accordance with the period load demand except that release may be delayed for a period not exceeding 60 seconds to insure stabilization of the flame. On cold startup, release may be delayed on units equipped with controls which prevent movement to high fire until a predetermined load is reached. Unless a medium is specified (see 6.2), combustion controls shall be electrically, pneumatically, or hydraulically operated. Air for pneumatic combustion controls shall be oil-free, suitably filtered, and properly dried. A means shall be provided to permit manual control of the firing rate, when desired. The system shall be so designed that it shall be impossible to put the fuel control on manual while the air control remains on automatic or vice versa and to fire more that one fuel at a time. A gas-oil selector switch shall be provided to select the appropriate fuel limits, main fuel trial for ignition timing, and select the appropriate main fuel valves. The switch shall have a center off position requiring a complete recycle of the programming control when changing fuels. All fuel limits shall be in the burner limit circuits and not in the main fuel valve circuit. On burners with High-Low-Off and modulating controls, the systems shall function to cause a normal shutdown in response to predetermined set point when the load demand drops below the minimum firing rate provided by the control system. Unless otherwise specified, (see 1.2, 6.1.1(d), 6.2, and table VI), the combustion control system shall be one of the following types.

3.11.1 Fixed rate, On-Off control. The fixed rate, On-Off control system (applicable to size 1 only) shall operate in response to load demand to cycle the burner on and off. A fixed relationship shall exist between the volume of air and fuel admitted to the burner. The air flow control damper shall be manually adjustable to permit change of the air/fuel ratio when required by firing conditions. After the main flame is proven, units with On-Off controls shall continue to fire until the load demand is satisfied.

3.11.2 Three position High-Low-Off control. The three position High-Low-Off control system (applicable to size 1 and 2 only) shall automatically adjust the flow of fuel and air to the burner, as required, to provide either high or low fire in accordance with the load demand.

3.11.3 Positioning (simple) control. The positioning control system shall permit the volumes of air and fuel delivered to the burner to be varied simultaneously and gradually in accordance with the load demand while maintaining the proper air/fuel ratio. The control system shall include an adjustable direct positioning mechanism to maintain the proper ratio between the air supply opening and the area of the fuel valve opening.

3.11.4 Parallel positioning control. Parallel positioning control with air/fuel ratio control shall include a master proportional controller which shall be connected to the load to sense the changes in load. The output of the controller shall transmit a signal simultaneously to the forced-draft damper and fuel valve operators. A separate ratio-adjustment controller which has been calibrated for the system to prevent a fuel-rich mixture shall be utilized in the line to the fuel valve and shall vary the signal for the fuel flow proportionally to the air flow. This action shall change the fuel flow to establish the optimum air/fuel ratio. A separate ratio-adjustment controller shall be used for each fuel. The parallel positioning control shall be capable of offsetting variations in pressure-drop characteristics in final control elements, fuel viscosity, air temperature, or barometric pressure. When switching between automatic and manual operation, the transfer

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shall be smooth. All controller adjustment and calibrations of fuel feed and air flow shall be made at the front of the control cabinet or panel with flush-mounted indicators provided to show adjustments and the results.

3.11.5 Parallel metering control. The parallel metering control system shall provide adequate means for automatic regulation of both fuel feed and air flow rates at the same time when a change in firing rate is initiated by the master controller. Actual rate of fuel and air flows shall be measured, and the signals linearized through signal processors that increase control system accuracy at low loads. The simultaneous modulation of the signals shall minimize the effects of surging, slugging, barometric conditions, ambient temperature, and similar variables to maintain the optimum furnace pressure and air/fuel ratio over the entire operating range. The metering system shall provide high-low limiting capability as follows:

a. On an increase in firing rate, the combustion air shall increase simultaneously with, or before, the burner fuel flow.
b. On a decrease in firing rate, fuel flow shall be reduced simultaneously with, or before, air flow.
c. Lack of combustion air due to fan stoppage and improper damper operation shall cause fuel flow to be reduced immediately to prevent a fuel-rich condition.

In balanced-draft furnace application, furnace pressure controls shall be required to keep the forced and induced-draft flow in balance. All controller adjustments and calibrations of fuel feed, air flow, air/fuel ratio, and furnace draft or pressure shall be made at the front of the control and instrument panel with flush-mounted indicators provided to show the degree of adjustment. The parallel metering system shall provide smooth operation when switching between automatic and manual operation. During load changes, the system shall minimize the increase in smoke density reading above the performance requirements of 3.8.2. Separate fuel flow transmitters and controllers shall be used with each fuel.

3.11.6 Metering with oxygen compensation control (trim). Oxygen compensation (trim) shall be provided in addition to the metering system of control specified in 3.11.5. The oxygen analyzer shall provide a feedback signal to the controller to insure the air flow corresponds to the load demand. The controlling system shall have the capability to bias the excess air/load curve. The air/fuel ratio controller shall not accept signals outside the preset limits. Compensation shall be made in the steady-state excess air/load curve for the variance in composition of the fuel to be burned. When specified (see 6.2 and 6.1(1), the parallel metering control with oxygen compensa- tion shall be equipped to measure and monitor the unburned combustible (carbon monoxide and hydrogen) to determine the minimum level of oxygen required and transmit a corrective action signal to the oxygen compensation control. The system shall use a compensated sensor to measure the level of combustible in the flue-gas containing some oxygen (0.5 percent by volume minimum). The system shall be equipped with an override control to activate an alarm. On closure of the alarm the controller shall override the oxygen compensation control signal to the actuator and cause and increase in the air signal until the alarm is satisfied (opens)(up to a maximum of $\pm$ 15 percent of the control position). The override correction shall operate in the same time control mode as the oxygen compensation control and be field adjustable.

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3.12 Burners and fuel trains.

3.12.1 Oil-fired units. Burners for oil-fired units with On-Off combustion controls shall be of the straight, mechanical, pressure-atomizing type, or low pressure, air-atomizing type. Unless otherwise specified (see 6.2), burners for units with High-Low-Off and modulating combustion controls shall be one of the following types.

a. Pressure-atomizing type of the return-flow or variable-nozzle design.
b. Air-atomizing type; compressor or pump furnished with burner.
c. Steam atomizing type for size 4 burners and larger with operating pressures of 100 psig and over. When specified (see 6.2), steam atomizing type shall not be acceptable.
d. Multiple-nozzle, or single nozzle-two pressure, straight, mechanical, pressure-atomizing or low pressure air-atomizing type for units with High-Low-Off controls only.

Burners may include such features as flame-retention an ultrasonic oil atomization. High-Low-Off and modulating oil burners shall have a burner turndown (minimum continuous safe firing rate), as specified (see 6.1.1(e), 6.2, and 6.6), but shall be not more than 33.3 percent of the maximum continuous firing rate. Oil fuel systems shall be equipped with the safety interlocks, limit switches, and control valves specified in figures 1, 2, 3, and 4, and denoted in table III and IV. In addition, the systems shall include a fuel oil pump, an oil strainer, relief valve(s), pressure gage with pressure gage cocks, and when required, a pressure regulating valve and oil preheating equipment. Oil burners shall produce a clean, stable, and efficient flame at all firing rates using the specified grade of fuel oil. The burner flame shape shall fit the furnace without impingement on or overheating of the furnace walls. Pressure atomizing oil burners of the return flow type shall be equipped with a return flow safety shutoff and a spring-loaded check valve to prevent return oil leakage on safety shutdown. The fuel system may operate on either the constant pressure or the constant differential principle provided positive control and adequate atomizing pressure is maintained at all firing rates. Units equipped with low air pressure atomizing burners shall be furnished with an air compressor or pump and associated controls and accessories mounted integrally with the unit. Steam atomizing burners shall require not more than 0.2 pound of steam per pound of oil burned at the maximum firing rate, and shall be equipped with pressure reducing and control valves, traps, and interlocks specified herein to assure that dry steam is available at the burner nozzle on light-off. Burners shall be equipped with constant differential valves or pumps, and with either compressed air supply or integral pressure atomizing starting burners for cold startup. Cold start pressure atomizing burners shall have approximately 30 percent of the maximum capacity of the main burner.

3.12.1.1 Fuel oil pump. The fuel oil pump shall be a motor-driven positive displacement or variable displacement metering type having sufficient capacity to deliver not less than 150 percent of the volume of the specified grade of fuel oil required at maximum rated burner capacity and 125 percent of the designed burner pressure measured at the burner tip when handling oil at the specified temperature from a suction lift of not less than 15 feet. For burners size 1 and 2 firing light oil, No. 2 or less, the fuel pump may be driven from the fan motor.

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3.12.1.2 Strainers. Duplex strainers, self-cleaning strainer, or simplex strainers as specified (see 6.2) having bypass provisions shall be installed at the fuel pump suction connection, and at such other locations as may be necessary to insure the delivery of clean oil to the burner regardless of any internal strainer.

3.12.1.3 Relief valves. Automatic oil pressure relief valves shall be provided on the discharge side of the fuel oil pump, the discharge side of the fuel oil heaters, and at such other locations as may be necessary to prevent excess pressure to the burner. Relief valves shall be complete with return piping to the fuel oil suction or return line.

3.12.1.4 Preheaters. When specified (see 6.2), preheating equipment shall be furnished in accordance with the following:

a. Electric preheaters shall be furnished on units firing No. 4 or No. 5 (light) fuel oil if required for proper atomization in low pressure, air atomizing systems.
b. Electric preheaters shall be furnished on units firing No. 6, Navy special, and when required, No. 5 (heavy) fuel oils for initial light-off of cold units. Preheaters shall have sufficient capacity for the firing rate occurring when the combustion controls are in the position for low-fire starts. Electric preheaters may be furnished in the same shell as the steam or water heaters.
c. Steam or hot water oil heaters, as specified (see 6.2), shall be furnished on units firing No. 6 and Navy special fuel oil, and when required by the type burners used, No. 5 (heavy) fuel oil.

The preheating equipment shall handle the applicable grade of fuel oil delivered to the preheater inlet connection at the temperature specified (see 6.2). When preheated oil is used, fuel oil systems shall be equipped with a temperature gage.

3.12.1.4.1 Electric preheaters. Heaters rated at least 5000 watts, and a nominal 240 volts (V), shall be controlled by a magnetic contactor having a magnetic coil operated by a thermostat. Heaters of smaller size rated at a nominal 240V, and less than 15 amperes per circuit, may be controlled by a line-voltage thermostat switch or by a magnetic contactor with a coil operated by a thermostat. The electric power circuits shall also contain approved means for disconnection, and shall be protected by fuses or circuit breakers. Separate circuits shall be provided for limit switches and control circuits shall not exceed 120V, shall be two-wire with one side grounded, and shall be protected by fuses or circuit breakers. When adequate heavy oil temperature is reached and maintained by the steam or hot water heater, the thermostatic control or limit switch shall cut off the electric heater unless the heater is used to minimize temperature fluctuations of the oil leaving the steam or water preheaters, or to provide additional heating for hot water heaters to maintain proper atomization temperatures.

3.12.1.4.2 Steam or hot water oil heater. The steam or hot water heater for oil shall be designed in accordance with section VIII of the ASME Code. Oil pressure drop through the heater shall not exceed 15 pounds-force per square inch. The heater shall be designed for boiler working pressures of steam and liquids at the corresponding steam or hot water temperature. The heater may be of the bare tube or extended surfaces type. The heater shall be

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equipped with an adjustable oil temperature regulator or aquastat and, when necessary, a steam pressure regulator, which will maintain the oil temperature within 5°F of the set temperature. The fuel oil pipes and valves shall be arranged to automatically recirculate hot oil through the piping system to the oil-control valve or safety-shutoff valve during stand by periods between firing cycles. The heating systems when used with a boiler shall include provisions for detecting or preventing oil leakage into the boiler. On steam heaters, means for visually detecting oil in the heater condensate line shall be provided. For hot water heaters, an auxiliary heating medium shall be interposed between the oil and boiler water. Alternate means for preventing entrance of oil into the boiler will be acceptable, subject to approval by the contracting officer.

3.12.2 Gas-fired units. Power burners for gas-fired units shall be of the partial premix or of the postmix type. Gas fuel systems shall be equipped with the safety interlocks, limit switches, and control valves specified in figures 1, 2, 3, and 4, and denoted in tables III and IV. The system shall include a drip leg and gas strainer installed ahead of the main pressure regulator. A pressure gage shall also be installed between the downstream safety-shutoff valve and the lubricated plug valve. Partial premix burners shall be provided with an integral primary air fan, and the design shall incorporate an effective means of preventing flame propagation back into the mixing chamber, or gas or air supply. The combustion air and gas supply shall be controlled to prevent gas from entering the burner until primary air is available, and to shut off the gas in case of primary air failure. When provided with the specified type of gas at the specified pressure, gas burners shall produce a stable flame which stays in one position relative to the burner. There shall be no blowoff from the burner face and no pulsating noise or vibration of the flame. Burner parts shall be protected against overheating. Fuel ports shall be machine made. High-Low-Off and modulating gas burners shall have a burner turndown, as specified (see 6.1.1(e), 6.2, and 6.6), but shall be not more than 33.3 percent of the maximum continuous firing rate. The burner flame shape shall fit the furnace without impingement on, or overheating of, furnace walls. Gas burners shall conform to ANSI Z83.3 for mechanical draft systems with interrupted igniter.

3.12.3 Combination gasand oil-fired units. A combination burner shall consist of an oil burner and a gas burner each conforming to the requirements of this specification for separate oil and gas burners for the sizes of burner specified herein. The two burners shall be constructed as an integral combination unit suitable for firing either fuel separately. Suitable provisions shall be incorporated for withdrawing, shielding, or otherwise preventing the oil burner from coking while firing gas. Except as otherwise specified herein, the fuel systems and fuel safety interlocks shall consist of separate systems and components for each fuel. The air-supply and boiler-limit interlocks shall be common to all fuels when practicable. The flame-failure system components may either be separate for each fuel or common to all fuels in accordance with the particular design. The burner shall be so designed that all controls will be transferred from one fuel to the other by a single manual selector switch furnished by the contractor. When the fuel is switched, the flame-failure controls, the combustion controls, and fuel-safety-limit controls shall be automatically switched to the appropriate fuel. On fuel change-over, the programming control system shall recycle on gas and light oil and shall recycle on heavy oil after the heavy oil is at the required temperature for proper atomization.

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3.12.4 Fuel trains. The burners shall have fuel trains conforming to UL 296, ANSI Z83.3 for power burners less than 12,500,000 Btu/hr, and NFPA No. 85A for burners greater than 12,500,000 Btu/hr. Oil piping shall withstand 1.5 times the fluid working pressure without rupture or leaking. Gas piping shall show a leakage of gas not to exceed 1/2 ounce per unit per year. When only one fuel is to be fired the fuel train for the fuel not fired may be deleted as well as the atomizing system. Only one oil atomizing system will normally be used. Ignition systems conforming to 3.13 shall be used. Fuel trains shall be as specified in table III, table IV and the following:

a. Size 1 burners (400,000 to 2,500,000 Btu/hr) shall be in accordance with figure I.
b. Size 2 burners (2,500,001 to 5,000,000 Btu/hr) shall be in accordance with figure 2.
c. Size 3 burners (5,000,001 to 12,500,000 Btu/hr) shall be in accordance with figure 3.
d. Size 4 and 5 burners (12,500,001 Btu/hr and above) shall be in accordance with figure 4.

3.12.5 Class 6 burners. Class 6 burners shall be in accordance with 3.12.2 and 3.12.4 and shall be capable of firing light oil, heavy oil, or gas. The burner design shall not permit the firing of more than one fuel at the same time. The fuel train associated with firing heavy oil shall incorporate a pre-heater, of the electrical, steam or steam-electric type, at the contractors option. Pre-heaters shall be installed on the discharge side of the pump and shall have the required by-pass piping to allow for firing light oil. The bypass piping shall have a 3-way supervisory valve at the inlet of the pre-heater and a supervisory valve at the outlet of the pre-heater. The pre-heater and the 3-way valve shall be up-stream of any strainer or filters. The fuel pump for class 6 burners shall be sized to handle the heaviest oil that is to be fired. A second 3-way supervisory valve shall be installed on the suction side of the pump to allow the firing of either light or heavy oil. A third 3-way supervisory valve shall be installed in the return line to direct any return oil to the proper storage tank. The supervisory valves shall be connected to the selector switch in such a manner as to prevent more than one oil being fired at the same time (see 6.1g.).

3.13 Ignition systems. Burners shall be equipped with ignition systems designed to supply sufficient ignition energy to provide positive ignition of all fuel discharged from the main burner under interlock light-off conditions. Ignition timing for flame establishing periods shall be as specified in table II. The ignition system shall be integrated with the programming control system specified herein. Gas igniter trains shall be equipped with a pressure regulating valve, relief valve, bleed valves, approved solenoid-operated safety shutoff valves, and lubricated manual shutoff valves in accordance with table III, table IV, and figures 1,2,3, and 4. Oil igniter trains for size 2 burners in (c) below shall include a strainer, (either separate or integral with the nozzle), filter and two solenoid-operated safety-shutoff valves. If the oil igniter system includes a pump, (separate from the main fuel oil pump), which shuts down in response to the safety control circuit, only one igniter safety shutoff valve will be required. Oil igniters shall be of the straight mechanical or low-pressure air atomizing type designed for firing No. 2 fuel oil. Ignition transformers shall be rated on the secondary side at not less than 6000V for gas-electric igniters and 10,000V for oil-electric igniters. A normally open valve, UL

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listed for gas service, equipped for automatic closing and spring opening shall vent the igniter gas piping to the atmosphere, (outside of the installation room) between the two safety-shutoff valves if the igniter is rated in excess of 120,000 Btu/hr. Igniter gas piping shall conform to ANSI CSD 1 as applicable. Burners shall be equipped with ignition systems conforming to the following requirements:

a. Except as otherwise specified herein, all igniters shall be the interrupted type with gas-electric ignition.
b. On size 1 burners firing No. 2 fuel oil, direct electric spark ignition may be provided.
c. When specified (see 6.1.1(h) and 6.2), direct spark ignition shall be furnished on oil-fired burners in lieu of gas-electric igniters on size 2 burners having High-Low-Off combustion control systems with 20 gallons per hour limit on light-off where gas is not available.
d. When specified (see 6.1.1(h) and 6.2), liquid petroleum (LP) gas shall be used for emergency backup for natural gas ignition. The installation shall be certified (see 6.3) to be in accordance with NFPA No. 54 and No. 58. The system shall be equipped with at least two pressure reductions to reduce the LP gas pressure to the nominal natural gas pressure. A three-way supervisory manual shutoff valve at the entrance of the natural gas ignition burner shall prevent firing of both fuels simultaneously and be arranged to cause the burner to recycle on changing fuels. The tank shall have UL label or listing mark or bare the ASME Code symbol mark and have a capacity in gallons, as specified (see 6.2). Detail drawings of the installation shall be submitted for approval (see 6.3).
e. When specified (see 6.1.1(h) and 6.2) light oil spark ignition ignitors shall be furnished to light the main flame on size 3, 4, and 5 oil fired burners. Ignitors and piping shall be in accordance with NFPA 85A

3.14 Control panel. The controls, including operating switches, indicating lights, gages, alarms, motor starters, fuses, and circuit elements of control systems shall be mounted on a single control panel or cabinet and shall be designed for a separate mounting not on the burner. Switches shall include burner start, manual-automatic combustion control, fuel oil heater, and fuel selector switches, as applicable. Colored indicating lights shall be provided and shall be activated when the interlock is satisfied for the control sequences herein. Unless otherwise specified (see 6.2), the number arrangement, and colors of the lights shall be as specified in table V and the lights shall be permanently identified. The control panel or cabinet or the individual elements shall be provided with NEMA ICS 6 type 12 or when specified (see 6.2) NEMA ICS 6 type 1A. The control panel shall be mounted on a solid base and not hung from the burner where it could be subjected to vibrations or be a safety hazard for the operator. All terminals requiring connection upon installation of the control panel shall be permanently identified in such a manner that correct connections can be easily made by reference to the instructions and wiring diagrams provided with the equipment. Terminal connections for control functions shall be segregated from power circuit terminals. The control panel shall be installed in a safe location with the burner visible from the operator's station to protect the operator from flareback. Burner and equipment controls including the flame and igniter

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failure safeguard controls shall be mounted in the control panel attached to the side of the equipment (right or left side when facing the equipment), for wall mounting, or for remote mounting in a free-standing control panel as specified (see 6.2). When a free-standing control panel is to be furnished construction details covering cabinet location, mounted controls wiring for remote starting, and instruments shall be as specified (see 6.2 and 6.3). The flame safeguard and burner controls shall not be mounted on the burner. When specified (see 6.2), a main circuit breaker shall be furnished and mounted either adjacent to the control panel or packaged separately for remote mounting at the site as specified herein.

3.14.1 Annunciator. When specified (see 6.2), an annunciator shall be provided for burners sizes 3 through 5. Unless otherwise specified, (see 6.2), the annunciator shall indicate all nonrecycling burner shutdowns specified in 3.10.6, including operational control alarms, and provide at least two spare stations or points in addition to those specified herein. The annunciator shall not require special or additional limit devices, or change sensitivity of limit switches, or annunciate without the limit device being actuated. The annunciator shall provide means to indicate the first-limit or off-limit condition causing burner shutdown and to sound an alarm. The alarm shall as a minimum be 10 decibles (DB) on the "A" scale above the ambient sound level. The ambient sound level at the installation shall be as specified (see 6.2). The annunciator shall also have a manual reset button and a capability to store information on additional causes of burner shutdown. Subsequent causes of burner shutdown will be indicated after the operator acknowledges the first burner shutdown. A dust-tight enclosure shall be provided in accordance with NEMA ICS 6 type 12 or when specified (see 6.2) NEMA ICS 6 type 1A. The annunciator shall include, but not be limited to, additional features and information required in Section 5 of ISA S18.1 as specified (see 6.2). Unless otherwise specified (see 6.2), the annunciator may either be a separate item, or an integral part of the control panel or burner control system.

3.14.2 Smoke density indicator and transmitter. When specified (see 6.2), an electronic opacity monitoring device with strip chart recorder and transmitter for stack mounting shall be provided to detect, indicate, alarm, and record visible emissions from the stack. The system shall conform to the requirements of the EPA Code of Federal Regulations Title 40, Part 60.344, Method 9, "Visible Emissions." A scale indicator shall be provided to display the output of the smoke density transmitter calibrated in percent opacity. The instrument need not have a clean air self-cleaning feature. The instrument shall operate in normal 120V (minimum 102V, maximum 132V), 60 hertz, and have internal electronic voltage regulation with power consumption of 20 amperes maximum. The smoke density indicator and transmitter locations shall be as specified (see 6.2).

3.15 Draft fan. The draft fan shall be for a pressurized or balanced draft system as specified (see 6.2); shall be integrally or separately mounted as specified (see 6.2); shall be constant speed or variable speed as specified (see 6.2); and shall be electric motor driven for direct or V-belt drive, as specified (see 6.2). The centrifugal or axial draft fan shall be rated for pressure, power, flow rate, speed of rotation, and efficiency in accordance with the test methods of AMCA 210. A manual control switch shall be provided to permit operation of the fan for purging the furnace prior to a cold start. Fan noise shall not exceed 90 DB on a "A" scale of a standard sound level

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meter at 3-foot stations at a height of 4.5 feet. The fan noise shall drop at least 6 DB when measured at a distance of 6 feet and 4.5 feet in height. All belt driven fans shall have safety guards that conform to ANSI B15.1 and adjustable motor mounts for belt tension adjustment and belt replacement. V-belts shall conform to ZZ-B-225. Sheaves shall conform to NEMA MG1 for minimum electric motor sheave diameter for a given horsepower motor. Motor sheaves shall be of the adjustable pitch type which provides a 20 percent speed variation, selected so that the specified capacity will be delivered with sheaves at approximately mid-position. V-belt drives shall be selected for continuous duty in accordance with Engineering Standards of RMA with a service factor of 1.5. Bearings for integral mounted fans shall be of the type normally furnished by the burner manufacturer. Bearings for separately mounted fans shall be precision antifriction bearings that meet the requirements of FF-B-171 for an average life of 100,000 hours; the self-aligning sheave type; or roller bearings mounted in suitable pillow blocks as specified (see 6.2). Oil reservoirs shall be equipped with an oil level sight gage and drain plug. The draft fans shall be sized for an output volume of combustion air or hot combustion flue gases sufficient for complete combustion at maximum rated burner output plus an additional 10 percent. Complete combustion shall be based on the percent of excess air necessary to obtain at least the minimum carbon dioxide flue-gas content listed in table I.

3.15.1 Pressurized draft system. Adequate corrections shall be included in the fan selection or design for maximum burner pressure drop, furnace pressure loss, combustion air temperature, and plant elevation to give a static pressure available at the flue stack outlet of the furnace in inches of water, as specified (see 6.2). Centrifugal fans shall have either backward or forward curved blades for the pressurized draft system.

3.15.2 Balanced draft system. The balanced draft system shall have a forced draft fan and an induced draft fan, designed as follows:

3.15.2.1 Forced draft fan. The static pressure required at the fan outlet shall be as specified (see 6.2), plus the maximum burner pressure drop to give a positive or a balanced (zero) pressure, as specified (see 6.2) at the inlet of the furnace. Centrifugal fans shall have either backward or forward curved blades for the forced draft fan.

3.15.2.2 Induced draft fan. The inlet and outlet static pressure required at the induced draft fan shall be as specified (see 6.2). The induced draft fan shall be designed for handling hot combustion flue gases at a maximum temperature, as specified (see 6.2), and shall have water cooled bearings when operating over 600°F centrifugal fans for induced draft service shall have radial-tip or forward-blade designs.

3.16 Electrical requirements.

3.16.1 Motors. Motors for the driven auxiliaries shall be constructed and rated in accordance with the applicable provisions of NEMA MG1. Motors shall be general purpose, continuous duty, ball-bearing, induction motors. The motors shall be designed for operation on a power supply having the specified characteristics (see 6.2). Unless otherwise specified (see 6.2) motors shall be furnished with drip-proof enclosures with encapsulated or sealed windings.

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3.16.2 Motor starters. Unless otherwise specified (see 6.2), motors less than 15 horsepower (hp) shall be provided with magnetic across-the-line starters with overload protection and manual-off-automatic control stations. Motors over 15 hp shall have reduced voltage starting with overload protection and manual-off-automatic control stations. Magnetic starters shall conform to the applicable requirements of NEMA ICS 1, ICS 2, and ICS 6, and shall have NEMA ICS 6 type 12 enclosures or when specified (see 6.2) NEMA ICS 6 type 1A enclosures.

3.16.3 Wiring and safety devices. All electrical wiring and circuit safety devices shall be suitable for operation at a minimum of 167°F and shall conform to UL 296 and NFPA 70. All control circuits shall be two wire, one side grounded, not to exceed nominal 120 voltage, and be protected with suitable fuses or circuit breakers. All safety interlocks and limit switches shall be wired in the hot side of the burner limit control circuit and not in any external circuits such as pilot or fuel valve circuits. When a free-standing control panel is required in 3.14, automatic sequence burners shall be wired for remote starting from the panel.

3.17 Lubrication. Unless otherwise specified (see 6.2), means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. All parts requiring lubrication shall be lubricated as specified in section 5. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high pressure lubricating equipment, 1,000 pounds-force square inch gage or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location.

3.18 Fungus resistance. When specified (see 6.2), electrical components and circuit elements, including terminal and circuit connections, shall be treated for moisture and fungus resistance in accordance with MIL-T-13867 as specified herein. Electrical components with a temperature rise of 40 degrees Centigrade (°C) or less shall be coated in accordance with type II of MIL-T-13867 with fungus resistant varnish conforming to MIL-V-173. Components with a temperature rise exceeding 40°C shall be coated in accordance with type III or IV of MIL-T-13867 with varnish conforming to MIL-I-24092 with grade and class of varnish dependent on the component, class of insulation above 40°C and the recommendations of the manufacturer of the equipment.

3.19 Electromagnetic interference suppression. When specified (see 6.2), the unit shall conform to the electromagnetic interference suppression requirements and test limits for class 3 group III equipment as specified in MIL-STD-461.

3.20 Lifting attachments. Suitable lifting attachments and tiedown devices shall be furnished in accordance with the manufacturer's standard practice except as specified herein. The attachments and tiedown devices shall have eyes not less than 3 inches in inside diameter, and plates to withstand any handling conditions encountered. The tiedown devices shall be located in accordance with the manufacturer's standard practice. Adequate tiedown devices may also be used for lifting purposes. Each attachment or tiedown device shall be of sufficient capacity to carry the total weight.

3.21 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2).

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Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.22 Technical manuals. The contractor shall provide technical manual(s) conforming to the Data Item Description(s) (DID) listed on the Contract Data Requirements List (CDRL) (DD Form 1423) (see 6.3).

3.23 Instruction plates. Unless otherwise specified, (see 6.2), the unit shall be equipped with instruction plates suitably located, describing procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment, and shall be securely affixed thereto with nonferrous screws or bolts of not less than 1/8-inch diameter. (see 6.3).

3.24 Identification marking. Identification shall be permanently and legibly marked directly on the burner or on a corrosion-resisting metal plate securely attached to the burner at the source of manufacturer. Identification shall include the manufacturer's model and serial number, name and trademark to be readily identifiable to the manufacturer. Electrical rating, type of flooring approved for mounting, installation clearances, and other information required under UL296, ANSI Z83.3, or NFPA No. 85A as applicable.

3.24.1 Military Marking. When specified, (see 6.2). An identification plate will be furnished by the contracting officer for each burner. The contractor shall stamp all necessary data in the blank spaces of the plate provided for that purpose, and securely affix a plate to each burner in a conspicuous place with brass screws or bolts not less than 1/8 inch in diameter. The applicable nomenclature contained in the contract item description shall be placed in the top blank.

3.25 Workmanship. The quality of workmanship shall be such as to produce burners that meet the requirements of this specification and standards prevailing among manufacturers normally producing this equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).
c. On-site inspection (see 4.2.3).

4.2.1 First article inspection. The first article inspection shall be performed on one burner when a first article is required (see 3.2 and 6.4). This inspection shall include the examination of 4.3, the tests of 4.5.1, 4.5.2, 4.5.3, 4.5.5, and when specified, the particulate emission test of 4.5.4 and the preproduction pack inspection of 4.6 (see 4.5.4, 4.6, and 6.2). The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.5.1, and 4.5.2, and the packaging inspection of 4.6.

4.2.3 On-site inspection. This inspection shall be in addition to the quality conformance inspection performed at the factory and shall be supplemental to the first article inspection, if such inspection is specified. On-site inspection shall consist of tests in 4.5.2., 4.5.3, 4.5.5, when applicable 4.5.4, and all examinations and tests deemed necessary by the acquisition activity to verify compliance with the requirements of this specification. On-site inspection shall be performed either by the burner manufacturer, or by the installing contractor, as specified (see 6.2). The manufacturer shall have the privilege of representation at tests performed by others. When the manufacturer is responsible for on-site tests, the detailed requirements and schedule for the test program will be as specified in the contract (see 6.2). In all cases, deficiencies revealed by on-site testing shall be corrected at the contractor's expense and any required retesting shall also be at the supplier's expense. This inspection shall be performed on the burner at the site after installation as specified herein.

4.3 Examination. Burners furnished under this specification shall be examined during production in accordance with the manufacturer's established quality control procedures, which shall assure compliance with all requirements of this specification that can be checked visually and with those elements of the manufacturer's drawings or specifications normally verified during the inspection process. Examination shall apply to all areas of contract performance including the receipt and identification of material, fabrication of the components, painting of parts, final assembly of the units including the installation and checkout of the control system and accessories. Quality control procedures shall be subject to surveillance by the acquisition agency or designated representatives thereof. Failure of the

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manufacturer to demonstrate that adequate quality control was applied to units submitted for acceptance by the Government may be cause for rejection of the units.

4.4 Standards compliance. The contractor shall make available to the contracting officer or his authorized representative evidence of compliance with the applicable standards cited in 3.4. The Government reserves the right to examine and test all burners to determine the validity of the certification.

4.5 Tests. Pressure and operational tests shall be performed on each burner supplied under a contract, including the first article sample, if furnished. Capacity, efficiency fire-tests, and auxiliary equipment tests shall be performed only on the first article sample. On burners rated up to 30 million Btu/hr input capacity, operational fire-tests of 4.5.2 shall be performed at a test installation provided by the burner manufacturer or performed at the location where the burner is installed at the option of the contractor. These tests shall be in addition to any manufacturers standard fire test procedure. Burners rated above 30 million Btu/hr shall be operationally fire-tested as a part of the on-site inspection (see 4.2.3).

4.5.1 Fuel system pressure test. The preassembled fuel system furnished integrally with the burner shall be checked for leakage during the fire testing of the burner. The halogen sniff test method shall be used to verify tightness of the gas system. The fuel oil system shall be hydrostatically tested in accordance with 3.12.4.

4.5.2 Operational fire-tests. Each burner shall demonstrate control and operational conformance to the requirements of this specification under varying load conditions, without On-Off cycling, that ranges from the specified capacity to the minimum burner turndown. All burners shall be tested for ability to cold start at the specified oil temperature (see 3.8 and 3.12.1.4) in accordance with the manufacturer's instructions. Burners that exhibit (1) excessive or unexplained loss of ignition, (2) nuisance shutdown due to faulty burner or control operation, (3) improper flame, (4) excessive carbon deposits, or (5) necessity for frequent or difficult adjustments shall be rejected. Operational tests shall be as specified in 4.5.2.1 through 4.5.2.6.

4.5.2.1 Sequencing. The burner shall start, operate, and stop in strict accordance with the specified operating sequence.

4.5.2.2 Flame failure. The operation of the flame failure control shall be verified by simulated flame and ignition failures. The igniter establishing period, the main-burner establishing period, flame-failure reaction time, and the safety-shutoff valve closing times shall be verified by stop watch (see table II and table IV).

4.5.2.2.1 Immunity to hot refractory. The burner shall be operated at high fire until the combustion chamber refractory reaches maximum temperature. The main fuel valve shall then be closed manually. The combustion safeguard shall drop out causing the safety shutoff valves to close within the specified flame-failure reaction time.

4.5.2.2.2 Minimum igniter test. The fuel supply to the igniter flame shall be gradually reduced to the point where the combustion safeguard begins

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to drop out (sense "no flame") but holds in until the main fuel valve opens. At this point of reduced igniter fuel supply, the igniter flame shall be capable of safely igniting the main burner. If the main fuel valve can be opened on an igniter flame of insufficient intensity to safely light the main flame, the burner shall be rejected.

4.5.2.3 Spark rejection test. With the fuel supply to both the igniter and main flame closed, the burner shall be operated into the ignition cycle. When the spark ignition comes on, the flame detector shall not be activated. If the flame detector can be activated, the burner shall be rejected.

4.5.2.4 Burner turndown. The specified burner turndown for High-Low-Off and modulating controls shall be verified by firing at the minimum firing rate required by the specified burner turndown (see 3.12.1 and 3.12.2).

4.5.2.5 Smoke tests. The smoke density reading shall be determined in accordance with ASTM D2156 and shall not exceed the smoke scale numbers at the percentages of rated load in table I.

4.5.2.6 Combustion controls. The accuracy range and smoothness of operation of the combustion controls shall be demonstrated by varying the load demand through the entire firing range required by the specified burner turndown, and in case of On-Off controls, by further varying the firing rate to require On-Off cycling. The control accuracy shall be as specified in 3.11.

4.5.3 Capacity and efficiency fire-tests. Input capacity specified in 3.8.1 shall be measured and recorded. Efficiency at the specified capacity of the burner shall be determined by an analysis of the flue gases. Performance shall be as specified in 3.8.2 and 3.11. Test runs shall be made at maximum capacity and for units with High-Low-Off at the lower limit of the burner turndown. Six samples of the combustion gases for each load setting shall be drawn evenly distributed across the face of the flue exit. If the burner is installed in a boiler the samples shall be drawn at the exit of the last gas pass. A variation in each flue gas value of over 2 percent may be an indication of air infiltration. Testing should be terminated until the cause of the variance has been identified and the problem corrected.

4.5.4 Tests for particulate and nitrogen emissions. When specified (see 6.2), tests for emissions of particulates and nitrogen shall be conducted and the method of test for particulate emissions shall be as specified (see 6.2). Tests shall be performed by the burner manufacturer, or by the installing contractor, as specified (see 6.2). The manufacturer shall have the privilege of representation at tests performed by others. The location of tests shall be as specified (see 6.2 and 6.3).

4.5.5 Auxiliary equipment tests. Draft fans, fuel oil heaters, fuel pumps, and electric motors shall be tested, when necessary, to determine compliance with the referenced specifications or standards. Standard approval symbols or certificates of compliance from certifying organizations may be accepted at the discretion of the contracting officer or his authorized representative. The operation of fans, fuel oil heaters, fuel pumps, and electric motors shall be closely observed during the operational tests for possible defects or nonconformance. The action of dampers, actuators, and valves shall be smooth without backlash.

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4.5.6 Electromagnetic interference suppression tests. To determine conformance to 3.19 the first article shall be subjected to the tests in accordance with MIL-STD-462. When suppressed to conform to 3.19, the manufacturer upon approval of the contracting officer may furnish a certification (see 6.3) in lieu of the test that the burner meets the requirements, together with a list of the suppression devices installed. The list shall be sufficiently detailed to allow visual determination that the devices are installed.

4.6 Packaging inspections. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of section 4 of MIL B-3180. The inspection shall consist of the quality conformance inspection and when specified (see 6.2), a preproduction pack shall be furnished for examination and test, and within the same time frame required (see 6.2).

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with MIL-B-3180 with the level of preservation and level of packing as specified (see 6.2).

6. NOTES

6.1 Intended use. The power burners are intended for heat generation with flame temperatures above 1600°F for use in steam and hot water boilers for power generation and space heating, for use in solid or liquid waste furnaces, and chemical process industries. The power burner to be selected should be determined in accordance with the availability or economics of the fuel supply and the intended application. Multi fuel burners should be specified herein in order to assure a reliable source of fuel. Equipment with installed superheaters may require a burner turndown as low as 10 percent of the maximum firing rate and may require a multi-burner installation which is not in the scope of this specification.

6.1.1 Selection factors. The following should be considered in the selection of the burner and the exercise of options herein:

a. With the present solid state programming control reliability, automatic (nonrecycling) burners should only be specified when automatic burners are not desired, or where full attendance at the plant is necessary for security reasons.
b. Single burner units should normally be specified for input capacities up to 100 million Btu/hr in a single furnace enclosure. The power burner should be sized to permit efficient burner operation under low load conditions.
c. Repetitive self-checking circuits have been included as standard because burners operating in occupied buildings or in hazardous locations require such safety controls (see 3.10.5.1).
d. The type of combustion control system should be selected in accordance with table VI based on burner input capacity of 3.8.1 (see 3.11 and 6.2).
e. Burner turndowns higher than those required or useful for the combustion controls should not be specified. Pressure atomizing burners are usually limited to a minimum firing rate of 33.3 percent of the maximum firing rate and are generally used for size

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1 and 2 burners. Most air atomizing burners specified herein and equipped with modulating controls, may have minimum firing rates of 20 to 25 percent of the maximum firing rates. Burners having input capacities above 30 million Btu/hr can be readily obtained, equipped with modulating controls with minimum firing rates of 16.7 percent of the maximum firing rate. Generally burner turndown can only be controlled by controlling the excess air and this will change with alternate fuels.

f. When gas burners are procured for stock and the conditions at the intended site are not known in advance, a gas pressure at the inlet to the gas piping of 2 psig should be specified for design purposes (see 6.2).

g. Normally heavy oil burners, above 2,500,000 Btu/hr input, (size 2), should be capable of burning fuel oil grades No. 2, No. 4, No. 5, and No. 6 without difficulty. The pump and motor drive selected for each burner should be sized based on firing the heaviest grade of oil because of the higher viscosity. When No. 5 and No. 6 oil is to be fired the burner should be provided with a pre-heater to reduce the oil viscosity. When it is required to fire more than one grade of oil, as in class 6 burners, separate fuel storage systems and fuel trains are required. Care should be exercised when more than one oil is to be used due to possible incompatibilities from different bases, (ie. paraffin and asphalt). The fuel train for firing No. 6 oil requires a pump and a heat exchanger (in addi- tional to the pre-heater at the burner) to heat the oil to the point it can be pumped. The fuel train shall incorporate a separate return line to the storage tank. Where temperatures drop below 40°F a suction stub heater installed in the storage tank will be required with heat traced oil lines to leep the No. 6 oil in the pumpable range.

h. Gas-electric igniters or direct spark ignition as specified herein are required for UL, FM, and Industrial Risk Insurance requirements. Oil igniters should be used only where gas is totally unobtainable or unfeasible for size 3 burners or larger. LP gas should be used as backup for the natural gas ignition system since it can be adjusted to fire through the same igniter (see 6.2).

i. If an oil burner is equipped with a single atomizer and if provisions for clearing the passages of the atomizer into the furnace are included, such clearing should be done after a normal shutdown with the igniter and fan operating under interlock conditions which proves the igniter, prevents main-fuel valves from being opened, and permits clearing valve to be opened only during clearing cycles.

j. A smoke density indicator should be specified on oil fired and combination burners over 5,000,000 Btu/hr input capacity (see 6.2).

k. If the burner is subject to environment regulation, each agency should consult with their environmental support office since regulation limits of particulate and emissions vary from city, state, and Federal governments (see 6.2).

l. On burners 10,000,000 and greater if excess air must be held below 4 percent on high fire to maximize combustion efficiency, oxygen trim control or oxygen compensation may be added to the control system (see 6.2).

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6.1.2 Other fuels. The performance requirements specified herein may be based on other fuels providing the quantity on-site is sufficient for test specified in section 4 and a guaranteed ultimate analysis, higher heat content, and oil viscosity in saybolt universal seconds (sus) are made available at time of testing the burners. Corrections between the values of the analysis in section 3 of this specification and the guaranteed analysis and heating value should be made in accordance with section 7 of ASME PTC-4.1. Referenced below are some of the most common used fuels with their average viscosity and higher heat content:

Fuel	Average viscosity	Average higher heat content
Light Oil		
Grade 2 and 4 ASTM D396	125 SUS	140,000 Btu/gal
JP-5 MIL-T-5624		
Diesel VV-F-800		
Heavy oil		
Grade 5 ASTM D396	300-750 SUS	150,000 Btu/gal
Navy Special MIL-F-859		
Grade 6 ASTM D396	1000 SUS	150,000 Btu/gal
	Specific gravity	
Natural gas	0.65	1075 Btu/cu ft
Manufactured gases	0.38	535 Btu/cu ft
Mixed gas	0.50	800 Btu/cu ft
Liquefied petroleum gas	1.55	2500 Btu/cu ft

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. The size of burner, class with type of fuel, control sequence, and combustion control system (see 1.2, 3.8, 3.11, 6.1.1(d).
c. If specifications and standards are different (see 2.1.1).
d. When first article is required for inspection and approval (see 3.2, 4.2.1, and 6.4).
e. When the burner is intended for conversion or replacement, submit the details of the installation (see 3.5.1, and 6.5).
f. When performance criteria shall be based on other than as specified: gas supply pressure at the site except for oil-fired units with direct-electric ignition or light-oil pilots) (see 3.8, and 6.1.1(f)).
g. The maximum continuous input capacity of the burner in Btu/hr (see 3.8.1 and 6.6.3).
h. When equipment shall be designed to burn fuel within emission limits: required emission limits of particulate nitrogen oxides and sulfur oxides and authority for regulations; when tests are required and method of testing; whether tests are to be conducted by the burner manufacturer or installing contractor and location of tests (see 3.8.2, 4.5.4, and 6.1.1(k)).
i. Type of combustion control required and medium of operation required, electric, pneumatic, and hydraulic (see 3.11, 6.1.1(d) and table VI).

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j. When parallel metering control with oxygen compensation shall be equipped to measure and monitor unburned combustibles (see 3.11.6 and 6.1.1(1)).
k. When oil burners with High-Low-Off or modulating controls shall be limited to a specific type or types (see 3.12.1).
l. When steam atomizing type oil burners are not acceptable (see 3.12.1).
m. Required burner turndown expressed as percentage of maximum continuous firing rate (see 3.12.1, and 3.12.2, 6.1.1(e) and 6.6.2).
n. Types of strainer(s) required (see 3.12.1.2).
o. When oil preheaters are required with types and temperature of the oil at each preheater inlet (see 3.12.1.4).
p. When direct spark ignition shall be furnished on size 2 burner with High-Low-Off combustion control system (see 3.13(c) and 6.1.1(h)).
q. When LP gas is required for emergency backup to the natural gas ignition system and size of tank (see 3.13(d), and 6.1.1(h)).
r. When arrangement of the color of the control panel indicating light are different, if NEMA IA encloser is acceptable, when cabinet is for mounting on the right or left of the equipment or for remote mounting in a fuel-standing control panel with details of its construction, location, mounted controls, instruments and wiring for remote starting (see 3.14).
s. When a main circuit breaker is to be furnished and the location of mounting (see 3.14).
t. When an annunciator is required for size 3 through size 5 burners and if different from as specified, when NEMA ICS 6 type IA enclosure is acceptable, additional features and information required in accordance with section 5, specifications of ISA 18.1, whether annunciator is to be a separate item or an integral part of the control panel or burner control system and ambient sound level at the installation site in DbA, (see 3.14.1).
u. When a smoke density indicator and transmitter is required and location of each. (see 3.14.2 and 6.1.1(j)).
v. Type of draft system required; required type of mounting for fan; direct or V-belt electric-motor-driven; type of speed, and speed ratio (if applicable) and type of bearings required for separate mounted fans (see 3.15).
w. For pressurized draft system, static pressure at the flue stack outlet of the boiler (see 3.15.1).
x. For balanced draft system, static pressure required at the forced draft fan outlet; the positive or balanced (zero) static pressure at the inlet of the furnace; the inlet and outlet static pressure required of the induced draft fan; and the maximum temperature of the flue gases (see 3.15.2.1 and 3.15.2.2).
y. Characteristics of the electrical power supply available at the site and if motor enclosures are different (see 3.16.1).
z. If motor starters are different and when required NEMA type IA enclosures. (see 3.16.2).
aa. If lubrication is different (see 3.17).
bb. When treatment for fungus resistance is required (see 3.18).
cc. When electromagnetic interference suppression is required (see 3.19).
dd. Required color of finished coat, (see 3.21).
ee. If instruction plates are different, (see 3.23).
ff. When military marking is required (see 3.24.1).

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gg. Whether on-site inspection will be performed by the burner manufacturer, or by the installing contractor; outline of tests if burner manufacturer will perform the tests, (see 4.2.3).
hh. When a preproduction pack is required and the time frame required for submission (see 4.2.1 and 4.6).
ii. Level of preservation and level of packing required (see 5.1).

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved DD Form 1664, Data Item Description (DID), and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph No.	Data requirements title	Applicable DID No.	Option
3.4.1.2, 4.5.6, and 3.13.(d)	Certificate of compliance	DI-E-2121	None
3.5.1, 3.13(d), 3.14	Installation drawings	DI-E-7031	None
3.22, 3.23	Publications, commercial	DI-M-24010A	None
4.5.4,3.4.1.1, and 3.4.1.2	Reports of Test	DI-T-2072	None

(DIDs related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L, Vol. II, Acquisition Management Systems and Data Requirements Control List. Copies of DIDs required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article (see 3.2 and 6.2).

6.5 Design instructions. The following design information should be submitted by the contracting officer to the contractor, when the burner is intended for conversion or replacement in a known or existing boiler (see 3.5.1 and 6.2).

a. Equipment make and model.
b. Input capacity rating of the equipment.
c. Dimensions of the combustion chamber and flue gas passages.
d. Plan dimensions of the face plate and the base, including height of base of the equipment and height from floor to center line of burner installation.
e. Type of fuel used in existing equipment and if it is to be continued to be used.

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The contract documents for the conversion or replacement burner should require the contractor to submit to the contracting officer for approval, in triplicate, layout drawings, tests of materials, and equipment necessary to meet the requirements of this specification.

6.6 Definitions. The following definitions are applicable for use by the contractor and acquisition agency in the interpretation and implementation of the requirements of this specification. Other terms used in this specification should be interpreted in accordance with the definitions of UL296, UL795, NFPA No. 85A, or ANSI Z83.3.

6.6.1 Approved. Conformance to the provisions specified under 3.4.

6.6.2 Burner turndown. The burner turndown shall be the minimum continuous safe firing rate which can be effected automatically by the combustion control system during the normal firing cycle without interruption or change of burner components. The burner turndown shall be expressed as a percentage of the maximum continuous firing rate.

6.6.3 Input capacity. For this specification, the input capacity of a burner is the volume of fuel consumed per hour, times the chemical heat in the fuel per volume (higher-heat value of the fuel per volume). The site elevation and temperature conditions shall be as specified in 3.8.

6.6.4 Nonproprietary controls. Control marketed by a supplier on a non exclusive right to sell.

6.6.5 Nonrecycling safety-shutdown. Stopping burner operation by shutting off all fuel and ignition energy to the furnace by means of a safety interlock or safety controls and requiring a manual reset (see 3.10.2(g)).

6.6.6 Power burner. A burner in which oil or gas is supplied individually along with air at pressures; for oil, line pressure; for gas, exceeding line pressure; and for air, exceeding atmospheric pressure, this added pressure being applied at the burner.

6.6.7 Recycling shutdown. Stopping of a burner operation by a normal shutdown when certain interlocks and limit switches have sensed a limit or off-limit condition and then allows the burner to automatically recycle when the condition corrects itself.

6.7 Cross-reference. The following cross-reference of classification applies.

MIL-B-18796D	MIL-B-18796E
One type	One type
size 1	size 1
size 2	size 2
size 3	size 3
size 4	size 4
Not designated	size 5

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Classes 1,2,3,4, and 5	Classes 1,2,3,4, and 5
Not designated	Class 6
Control sequence I	Control sequence IA
Not designated	Control sequence IB
Control sequence II	Control sequence II
Control sequence IV	Control sequence IV
Combustion Control System	Combustion Control System
On-Off	On-Off
High-Low-Off	High-Low-Off
Positioning (simple)	Positioning (simple)
Parallel positioning with air fuel ratio control	Parallel positioning with air fuel ratio control
Parallel metering with Lead-lag capability	Parallel metering with Lead-lag capability
Metering with oxygen compensation	Metering with oxygen compensation

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:

Army - ME
Navy - YD
Air Force - 99

Review Activity:

Air Force - 84

User Activity:

Army - CE
Navy - MC

Preparing Activity:

Navy - YD

(Project 4530-0037)

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TABLE I. Requirements for Combustion

Maximum Burner Capacity (Input) From BTUh (X 1001)	Maximum Burner Capacity (Input) To BTUh (X 1000)	Type of Combustion Control	Percent Rated Load	Fuels: Natural Gas Percent CO_2 Minimum Note 1	Fuels: Light Oil No.2, JP-5 Diesel Percent CO_2 Minimum Note 2	Fuels: Medium Oil No. 4, NSFO Percent CO_2 Minimum Note 2	Fuels: Heavy Oil No. 5 and 6 Percent CO_2 Minimum Note 2
400	2,500	"On-Off"	100	9.5	12.0	12.5	Not Applicable
2,500	5,000	"Hi-Lo-Off" or "Modulating"	40	8.5	10.5	11.0	12.0
			70	9.0	11.5	12.0	12.6
			100	10.0	12.5	13.0	13.6
5,000	12,500	"Modulating"	40	8.5	10.5	11.0	12.0
			70	9.0	11.5	12.0	12.6
			100	10.0	12.5	13.0	13.6
12,000	30,000	"Modulating"	40	8.5	10.5	11.0	12.0
			70	9.0	11.5	12.0	12.6
			100	10.0	12.5	13.0	13.6
30,000	Above	"Modulating"	40	8.5	10.5	11.0	12.0
			70	9.0	11.5	12.0	12.6
			100	10.0	12.5	13.0	13.6
Average Combustion Efficiency (Percent)				80	81	81	81
Equivalent Bacharach Smoke Scale (Maximum)				0	2.0	3.0	3.0

Note 1 Natural gas units rated below 30,000,000 Btu/hr input capacity shall meet UL 795 requirements for carbon monoxide. Units rated above 30,000,000 Btu/hr inpurt capacity shall have no more than 0.05 percent carbon monoxide.

Note 2 All oil burning units shall have no more than 0.05 percent carbon monoxide.

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TABLE II. Programming control.

Burner size	Control sequence	Reference paragraph	Kind of igniter and ignition	Grade fuel (see 1.2, 3.8, and 6.1.2	Maximum Establishing periods Seconds Igniter	Mainflame
(1) 400,000 to 2,500,000 Btuh	IA, Automatic system (recycling)	3.10.1	Spark ignition No igniter	Light oil only	--	10 (see 3.10.1 (d))
	IB, Automatic system (recycling)	3.10.2	Interrupted proved igniter	Gas, light oil or gas and light oil combination	10	10 (see 3.10.2 (c))
(2,3) 2,500,000 to 12,500,000 Btuh	IB, Automatic system (recycling)	3.10.2	Interrupted proved igniter	Gas, light oil, gas-oil light combination all fuels	10	10 (see 3.10.2 (c))
	II, Automatic system (nonrecycling)	3.10.3	Interrupted proved igniter			
(4,5) over 12,500,000 Btuh	IB, Automatic system (recycling)	3.10.2	Interrupted proved igniter	Oil, Grades No. 4 through No.6	10	15 (see 3.10.2 (c))
	II, Automatic system (nonrecycling)	3.10.3	Interrupted proved igniter			
	IV, (Gas), Supervised manual system	3.10.4	Interrupted proved igniter	Note 1: All fuels in this box may be used with all control sequences to the left		
	IV, (Oil), Supervised manual system	3.10.4	Interrupted proved igniter			
	IV, (Gas-Oil), Supervised Manual system	3.10.4	Interrupted proved igniter			

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TABLE III. Identification, descriptions, and requirements for safety interlocks and limit switches.

Item No.	Description	Type of burner shutdown required
	See figures, 1, 2, 3, and 4	
	SAFETY INTERLOCKS AND LIMIT SWITCHES	
SC-3	Flame scanner furnished as part of the flame failure control system (see 3.10.5).	Nonrecycling
SC-5H	Gas supply, high pressure cut out switch manual reset type.	Nonrecycling
SC-5L	Gas supply, low pressure cut out switch manual reset type.	Nonrecycling
SC-7-1	Wind box pressure switch to monitor air flow for purge and firing cycles: To be furnished on all sizes of units, whether or not unit is equipped with centrifugal switch (fan motor interlock).	Nonrecycling
SC-7-2	Fan damper position switch to be furnished when fan damper position for purge is other than at light-off position.	Nonrecycling
SC-8	Closed position interlock on GB-4 and OB-4 (if requred)(valve seal overtravels to actuate an interlock).	Nonrecycling
SC-9	Light-off position interlock (low fire start on Modulating Control OC-10) to prove damper and fuel flow control valves are in low-fire position for light-off; not required on units with On-Off combustion control. Interlock to be part of or integrated with combustion control system.	Stops programer cycle until low fire position proven.
SC-10	Atomizing medium pressure interlock switch for burners with air- or steam atomizing burners, to cause nonrecycling safety shut-down on startup or during the firing cycle if the atomizing medium falls below a predetermined pressure for safe combustion.	Nonrecycling
SC-10-3	Atomizing medium supply pressure interlock switch for burners with air- or steam atomizing burners to prevent startup without adequate medium pressure.	May recycle
SC-11	Low oil temperature interlock for units with oil preheaters. Installed in a different tapping than the operating control. Set lower than OB-5.	Nonrecycling

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TABLE III. Identification, descriptions, and requirements for safety interlocks and limit switches. (Cont'd)

Item No.	Description	Type of burner shutdown required
	See figures, 1, 2, 3, and 4	
SC-12	Low oil supply pressure interlock; not required on light oil burners when the oil pump is integral with the burner. Switch to be manual reset type.	Nonrecycling
	NOTE: See table IV for additional interlocks.	

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TABLE IV. Identification, description, and requirements of atomizing medium fuel, igniter burner and air systems and operation control and instruments.

Item No.	Description
	See figures 1, 2, 3, and 4
	ATOMIZING MEDIUM SYSTEM NOTE 1
ATS-1	Atomizing prsssure gage, manufacturer's standard.
ATS-3	Atomizing medium shut-off valve solenoid operated may be omitted if an air/oil metering device is used.
ATS-4	Atomizing medium differential pressure control (if required).
ATS-5	Atomizing medium supply strainer manufacturer's standard.
ATS-6	Atomizing medium supply trap manufacturer's standard.
ATS-7	Atomizing steam supply check valve.
ATS-7A	Atomizing air supply check valve.
ATS-8	Atomizing steam supply shut-off valve.
ATS-8A	Atomizing air supply shut-off valve.
	Note 1. Gages and valves shall be suitable for the atomizing medium and pressure used.
	GAS SUPPLY SYSTEM
GS-1	Drip leg.
GS-2	Manual plug cock.
GS-3	Gas supply pressure reducing valve. Suitable for line pressure upstream (see 3.8).
GS-4	Manual gas supply shut-off valve.
GS-5	Gas supply pressure gage.
GS-6	Gas supply pressure gage cock.
GS-7	Gas cleaner.
GS-8	Gas relief valve piped and mounted to outside atmosphere.
GS-9	Nonelectric valve held open by a fusible link designed to close the valve at 165° F.

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TABLE IV. Identification, description, and requirements of atomizing medium fuel, igniter burner and air systems and operation control and instruments. (Cont'd)

Item No.	Description
	See figures 1, 2, 3, and 4
	AIR SYSTEM
AS-1	Forced draft fan.
AS-2	Forced draft fan motor.
AS-3	Forced draft fan control damper at inlet or outlet.
	IGNITER (PILOT) SYSTEM
P-2	Two solenoid-operated, safety shutoff valves, for Fig. 4, spring closing (NC) rated to close at a pressure higher than line pressure upstream from the pressure regulator P-4 and shall be 5 psig minimum approved for gas service.
P-2A	Solenoid-operated, safety shutoff valve for figures 1, 2, and 3, spring closing (NC) rated to close at at pressure higher than line pressure upstream from the pressure regulator and shall be 5 psig minimum approved for gas service.
P-3	Atmosphere vent valve, normally open, approved for gas service, equipped for automatic closing on a line pressure upstream higher than line pressure upstream from the pressure regulator P-4, required when total input to igniter is in excess of 120,000 Btu/hr, vented to atmosphere outside of building.
P-4	Gas pressure regulating valve operational suitable for line pressure upstream (see 3.8).
P-6	Manual plug cock.
P-8	Leak test valve for leakage test connection downstream of safety shut-off valves.
P-8-2	Leak test valve for leakage test connection between safety shut-off valves.
P-8-3	Leak test valve for leakage test connection downstream of the vent valve (P-3), is required only when vent valve (P-3) is used (see item P-3).
P-9	For igniters on figure 4 which use less than 120,000 Btu/hr a supervisory cock approved for gas service with double pole double throw (DPDT) switch and a separate DPDT power test switch.

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TABLE IV. Identification, description, and requirements of atomizing medium fuel, igniter burner and air systems and operation control and instruments. (Cont'd)

Item No.	Description
	See figures 1, 2, 3, and 4
P-9A	For igniters greater than 120,000 Btu/hr, a manual cock is used instead of P-9 when vent valve (P-3) is required.
P-10	Pressure gage, manufacturer's standard with cock valve.
P-11	Igniter low pressure switch, to be manual reset type, to cause a nonrecycling shutdown.
P-12	Igniter high pressure switch to be manual reset type, to cause a nonrecycling shutdown required only when total input to igniter is in excess of 120,000 Btu/hr.
P-13	Supervisory cock approved for gas service downstream of the vent valve (P-3), and leakage test connection (P-8-3) with DPDT switch, required only when vent valve (P-3) is used. (See item P-3).
	GAS BURNER SYSTEM
GB-1	Manual plug cock.
GB-2	Gas burner pressure gage manufacturer's standard.
GB-3	Gas burner pressure gage cock.
GB-4	For Sequence IB - Automatic (recycling) units, figures 2, 3, and 4, two approved motorized safety shutoff valves, normally closed, electric reset and holding. For figures 3, and 4 downstream valve shall have closed position interlock (SC-8). Valves shall be spring closing and rated to close at a pressure higher than line pressure upstream from the pressure regulator (GS-3) and shall be 5 psig minimum (see 3.8). Valve closing time shall not exceed 1 second on all burners.
GB-4A	For Sequence II - Automatic (nonrecycling) and Sequence IV, Supervised manual burners. One approved manual reset, normally closed safety shutoff valve, electric solenoid holding with closed position interlock (valve seal overtravels to actuate an interlock (SC-8)) located in the downstream position and one approved, motorized, safety shutoff valve normally closed, electric reset and holding located upstream from manual reset valve. Valve shall be spring closing and rated to close at a pressure higher than line pressure upstream from the pressure regulator (GS-3) and shall be 5 psig minimum (see 3.8). Valve closing time shall not exceed 1 second on all burners. These valves shall be used instead of GB-4 valves.

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TABLE IV. Identification, description, and requirements of atomizing medium fuel, igniter burner and air systems and operation control and instruments. (Cont'd)

Item No.	Description See figures 1, 2, 3, and 4
GB-4B	For sequence IB, Figure 1, - Automatic (recycling) units, safety shutoff valve, normally closed, electric reset and holding, solenoid operated, valve to be spring closing and rated to close at a pressure higher than line pressure upstream from the pressure regulator (GS-3) and shall be 5 psig minimum, approved for gas service (see 3.8), valve closing time shall not exceed one second.
GB-5	Vent valve, general purpose, approved for gas service, normally open, full ported electrically-operated solenoid valve connected in a manner to vent piping between safety-shutoff valves whenever valves are closed. Vented to atmosphere outside of building.
GB-6	Gas fuel control valve to control fuel flow rate for high-low-off and modulating combustion control systems (see 3.11), approved for gas service.
GB-6A	General purpose, solenoid valve for operational control on units with On-Off controls approved for gas service and shall be used instead of GB-6 valve.
GB-8-2	Leakage test connection downstream of safety shutoff valves.
GB-8-3	Leakage test connection between safety shutoff valves.
GB-8-4	Leak test valve for leakage test connection downstream of vent valves (GB-5).
GB-9	Supervisory cock approved for gas service downstream of the vent valve (GB-5) and leakage test connection (GB-8-4) with DPDT switch.
	OIL BURNER SYSTEM
OB-1	Manual oil shutoff valve.
OB-2	Oil burner pressure gage manufacturer's standard.
OB-3	Oil burner pressure gage cock.
OB-4	For sequence IA and IB - Automatic (recycling), burners, a solenoid operated, normally-closed, safety-shutoff valve, valve to be electric reset and holding. Two valves required on Figure 4. Downstream valve shall have closed position interlock (SC-8). Valves shall be spring closing. Safety-shutoff valves shall be UL listed or FM approved for the applicable oil service, and shall have a valve-closing time not to exceed 1 second.

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TABLE IV. Identification, description, and requirements of atomizing medium fuel, igniter burner and air systems and operation control and instruments. (Cont'd)

Item No.	Description
	See figures 1, 2, 3, and 4
OB-4A	For sequence II - Automatic (nonrecycling) and sequence IV - supervised manual burners. One approved manual reset, normally-closed, safety shutoff valve, electric solenoid holding with closed position interlock (SC-8) located in the downstream position, and one approved solenoid operated, safety-shutoff valve, normally closed, electric reset and holding located upstream from manual reset valve. Valves shall be spring closing. Safety-shutoff valves shall be UL listing or FM approved for oil service, and shall have a valve-closing time not to exceed 1 second. On units with On-Off combustion controls, the electric-reset safety-shutoff valve may serve as the operation valve in lieu of item OB-6. These valves shall be used instead of OB-4 valves.
OB-5	Oil temperature gage, with separate high and low temperature switches to limit temperature and sound an alarm and activate a red indicating light. High limit to cause a safety shutdown. The low limit to stop fuel firing to allow circulation to reestablish oil temperature, burner shall recycle. Limit switches and temperature gage may be omitted for light oils which do not require heating. Gage shall be UL listed.
OB-6	Oil control valve or metering device to control fuel flow rate for High-Low-Off and modulating combustion control system (see 3.11).
OB-7	Oil supply pressure gage, manufacturer's standard.
OB-8	Oil supply pressure gage cock.
OB-9	Oil strainer, manufacturer's standard.
OB-10	Check valve when oil recirculation system is used. Manufacturer's standard.
OB-11	Nonelectric valve held open by a fusible link designed to close the valve at 165° F.

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TABLE IV. Identification, description, and requirements of atomizing medium fuel, igniter burner and air systems and operation control and instruments. (Cont'd)

Item No.	Description
	See figures 1, 2, 3, and 4
OB-12	For sequence IA and IB for figures 1, 2, and 3, Automatic (recycling), burners, safety shutoff three-way valve solenoid operated normally closed, valve to be electric reset and holding, valve to be spring closing, shall be UL listed or FM approved for the applicable oil service, and shall have a valve closing time not to exceed one second. On sequence IA, On-Off Combustion control this valve shall serve as the operational valve.
OB-13	Oil pressure relief valve, adjustable, sized for the application.
	OPERATION CONTROLS AND INSTRUMENTS
OC-7	Combustion control system (drive unit) (see 3.11).
OC-10	Modulating control-low fire start positioner (see 3.11).

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TABLE V. Arrangement and colors of indication lights for control panel.

Operation	Color
Low water	Red
Safety lockout	Red
Atomization, medium, when used	Yellow
Auto control	Green
Power on	Blue

TABLE VI. Recommended selection of combustion control systems based on input capacities.

Input capacities thousand Btu/hr all fuels	Type of control	Reference paragraph
400 - 2,500	On-Off	3.11.1
2,000 - 5,000	High-Low-Off	3.11.2
3,000 - 30,000	Positioning (simple)	3.11.3
15,000 - above	Parallel positioning with air/fuel ratio control	3.11.4
	Parallel metering	3.11.5
20,000 - above	Parallel metering with oxygen compensation (trim)	3.11.6

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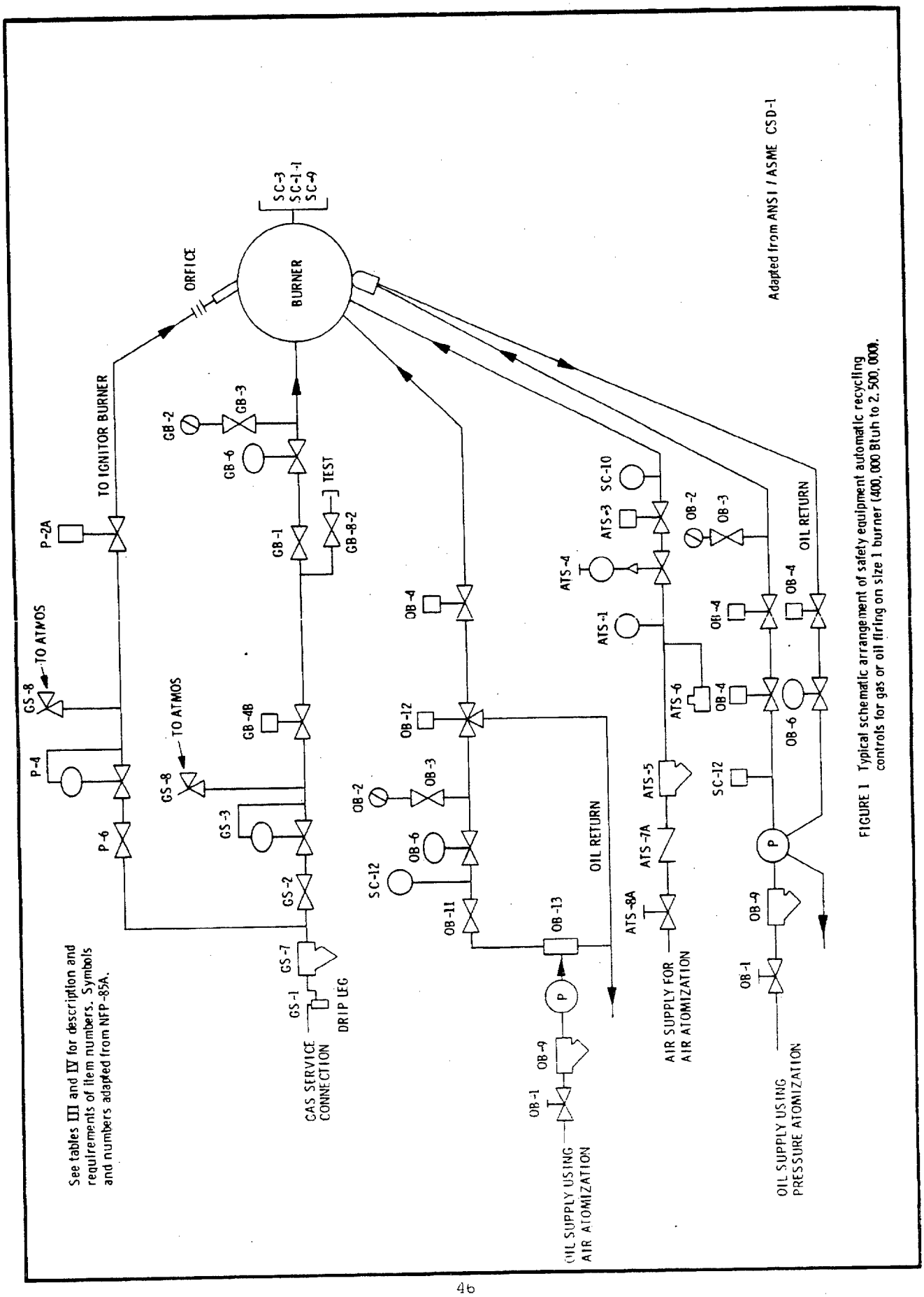


FIGURE 1 Typical schematic arrangement of safety equipment automatic recycling controls for gas or oil firing on size 1 burner (400, 000 Btuh to 2, 500, 000).

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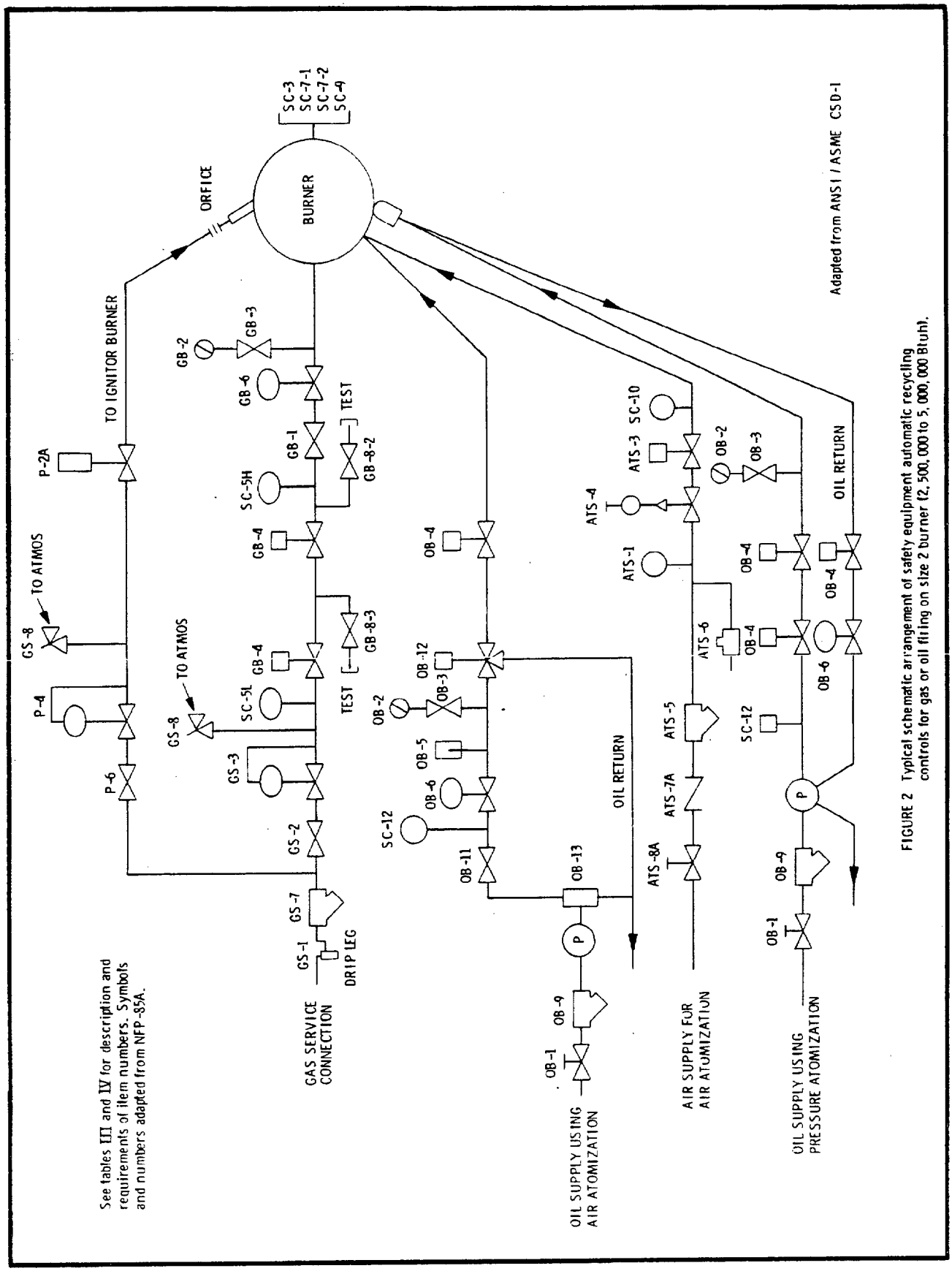


FIGURE 2 Typical schematic arrangement of safety equipment automatic recycling controls for gas or oil firing on size 2 burner (2,500,000 to 5,000,000 Btuh).

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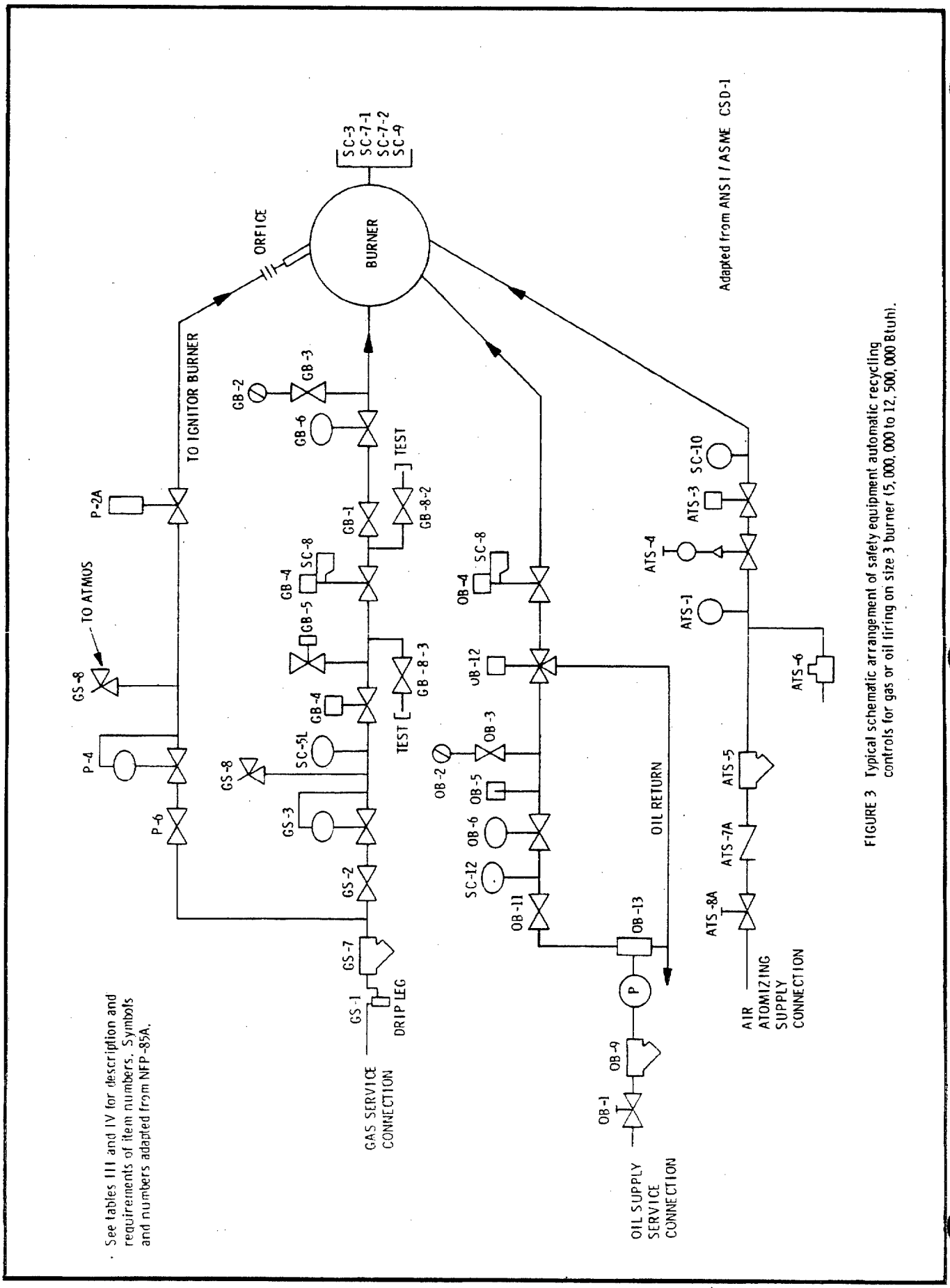


FIGURE 3 Typical schematic arrangement of safety equipment automatic recycling controls for gas or oil firing on size 3 burner (5,000,000 to 12,500,000 Btuh).

* See tables III and IV for description and requirements of item numbers. Symbols and numbers adapted from NFP-85A.

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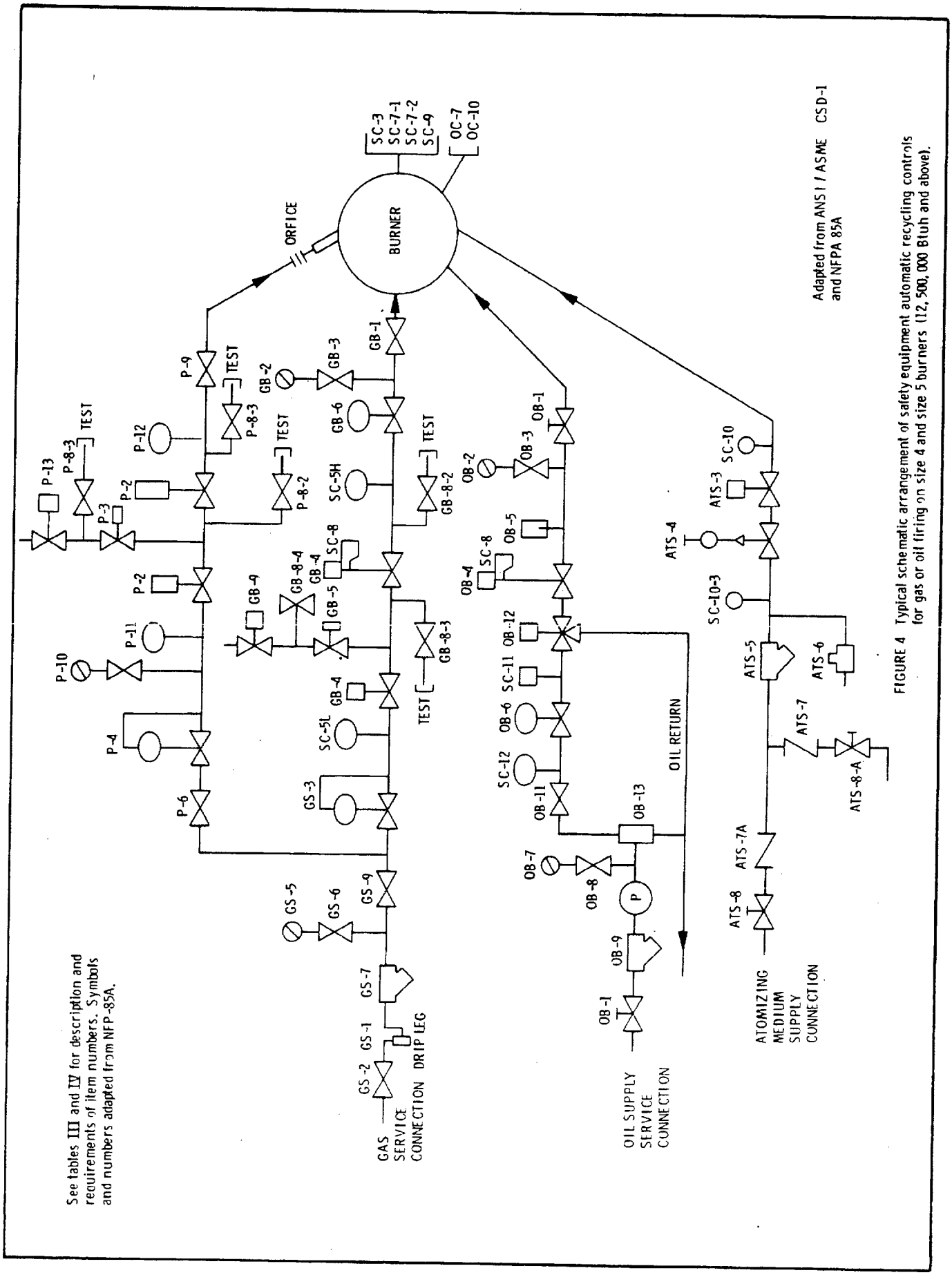


FIGURE 4 Typical schematic arrangement of safety equipment automatic recycling controls for gas or oil firing on size 4 and size 5 burners (12,500,000 Btuh and above).

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